

The Strategy Library

An encyclopedic reference of 50 systematic trading strategies, from RSI mean reversion to PEAD, Renko, and the volatility risk premium. Every entry: how it works, where the edge comes from, the regimes it lives and dies in, the settings to configure, how to size and risk it, and a Claude Code prompt to build it on your bot template.

50 Strategies

10 Families

Standalone & Regime-Gated

Config + Sizing + Build Prompt

Beginner to Quant

aipathways.io · Pairs with the automated trading bot guides and the regime-trader template

Every strategy maps onto the BaseStrategy / PairStrategy interface and the HMM regime engine.

How to use this library

This is a reference, not a course. Skim the family dividers to find a category that fits what you want to do, then read the individual strategies. Each one is self-contained and ends with a Claude Code prompt that builds it on your existing bot template, so you can go from reading about a strategy to backtesting it in one paste.

These are starting templates, not plug-and-play systems

Every strategy here is a starting point you configure to your own plan, not a finished system to run as-is. The defaults in each configuration table and build prompt are reasonable starting values, not recommendations for your account. You set the position sizing, the stop and max loss per trade, and the strategy settings based on your capital, your risk tolerance, and your own testing. Each build prompt is written to expose those as parameters and remind you to set them. Backtest on your own data before any of this touches real money.

Four dimensions of fit: regime, time, day, and asset class

A strategy working is not just about the signal. Each entry tells you four things about where it fits. Regime is the best and worst market state, usually tied to your HMM engine. Time of day covers whether it is a daily-close strategy or an intraday one, and when in the session it works. Day and calendar covers any weekday or calendar-window dependence. Asset class tells you which instruments it suits, since a rule built for index ETFs often fails on single small-caps or commodities. Match all four to the strategy before you run it.

Standalone vs regime-gated

Most strategies here can run on their own, but nearly all of them work better when gated by your HMM regime engine: mean reversion in calm regimes, trend in trending ones, premium selling switched off in crashes. Each entry lists its best regime and the regime to avoid. The Regime & Allocation family at the end is the connective tissue that decides how much of each strategy you run and when.

How each strategy is laid out

How it works and where the edge comes from explain the idea and the source of returns. When to use it covers standalone vs regime-gated. The regime boxes, the time-day-asset fit, the timeframe, and the assets tell you where it belongs. The configuration table lists the parameters and starting values. Risk and sizing is where you apply your own rules. Watch out for and evidence cover the failure modes and how well documented the edge is, including how much it has decayed since first published.

Ten families of strategy

The library is organized into ten families, ordered roughly from the most reliable and beginner-friendly to the most advanced and dangerous. Mean reversion and trend come first because they are the workhorses. Volatility, pairs, and the niche charting and microstructure families come later and demand more care. The allocation family at the end wraps around everything.

1	Mean Reversion (8)	Bet that price snaps back to an average after stretching too far.
2	Trend & Momentum (7)	Bet that what is moving keeps moving.
3	Cross-Sectional / Rotation (5)	Rank a basket of assets against each other and hold the leaders, rotate out of laggards.
4	Volatility & Options (5)	Trade volatility itself, or harvest the premium that option sellers collect on average.
5	Event & Calendar (6)	Trade around scheduled events and calendar effects: earnings, index rebalances, month-end, FOMC.
6	Pairs & Stat-Arb (4)	Trade the spread between two related assets rather than direction.
7	Alternative Charting (4)	Reframe price using bricks, ranges, or averaged candles to strip out noise.
8	Microstructure & Intraday (3)	Short-horizon strategies that exploit the structure of the trading day: the opening range, order-flow imbalance, gaps.
9	Seasonality & Macro (4)	Calendar-season and macro-driven tilts: sell-in-May, the turn-of-year effect, sector seasonality, carry.
10	Regime & Allocation (4)	Not signals at all, but rules for how much to hold and when.

Each family opens with its own divider page listing every strategy in it, followed by the full entries.

DISCLAIMER

Disclaimer

This document is educational. It is not financial advice, and nothing here is a recommendation to trade any strategy or instrument. Brendan and AI Pathways are not your financial advisor.

Every strategy in this library can lose money. The documented edges come from academic and practitioner research, and they are presented with their weaknesses and decay, not as guarantees. Past performance, backtested or live, does not predict future results. Many of these strategies were more profitable in their original studies than they are today.

Several strategies here, in particular short volatility, the volatility risk premium, and leveraged allocation, carry the risk of large and sudden losses that can exceed typical expectations. The risk warnings on those entries are literal, not boilerplate. Trade small, paper trade first, and do not deploy real money on a strategy you have not tested on your own data and cannot risk-manage.

You are responsible for your own trading decisions, your own risk, and your own capital.

Mean Reversion

Bet that price snaps back to an average after stretching too far. These are the most reliable category out-of-sample in broad testing, but they get run over in strong trends and crashes. Almost all of them want a calm or range-bound regime and a hard stop for the times the rubber band does not snap back.

MR-01 RSI-2 Mean Reversion

MR-02 Bollinger Band Reversion

MR-03 Z-Score Reversion

MR-04 VWAP Reversion (Intraday)

MR-05 Gap Fade

MR-06 Internal Bar Strength (IBS)

MR-07 Overnight / Close-to-Open Reversion

MR-08 Keltner Channel Reversion

MR-01

RSI-2 Mean Reversion

Larry Connors' classic: buy short-term oversold inside a long-term uptrend.

HOW IT WORKS

On a daily chart, only consider longs when price is above its 200-day moving average (the long-term trend is up). Buy when the 2-period RSI drops below an oversold threshold (the short-term pullback). Exit when price closes back above a short moving average (e.g. the 5-day) or RSI rises above a middle level. It is a pullback-buyer inside an uptrend, not a bottom-picker in a downtrend.

WHERE THE EDGE COMES FROM

Short-term overreaction to selling inside an established uptrend. Buyers who panic-sell a dip in a rising market tend to be wrong over a few days.

WHEN TO USE IT

Standalone on liquid index ETFs and large caps. Pairs extremely well with a regime filter: only take signals in a bull or low-vol regime, skip entirely in crashes.

BEST REGIME

Bull, low-volatility, range-bound-up

AVOID IN

Crash, sustained bear (the 200-day filter handles most of this, but a regime gate adds protection)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Signal computed on the daily close; act on the next open or the close. Not an intraday strategy.

DAY / CALENDAR

No strong day-of-week bias. Some find Monday/Tuesday dips revert a touch more often, but do not over-tune to it.

ASSET CLASS

Equity index ETFs and large-cap stocks. Avoid single small-caps (news swamps the signal) and do not apply to trending commodities or FX where the 200-day-up filter rarely lines up with clean reversion.

TIMEFRAME

Daily; holding 2 to 6 days

ASSETS

Index ETFs (SPY, QQQ), liquid large caps

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>rsi_period</code>	2	The short RSI. 2 is the classic; 3 to 4 is slightly slower.
<code>rsi_entry</code>	10 (aggressive) / 5 (very aggressive)	Lower = fewer, deeper signals. Some use <15 for more trades.
<code>trend_ma</code>	200-day SMA	Long-term trend filter. Only go long above it.
<code>exit_ma</code>	5-day SMA	Exit when price closes back above this, or use RSI>50.
<code>max_hold</code>	10 days	Time stop so a failed signal does not become a bag-hold.

RISK & SIZING — SET TO YOUR OWN PLAN

Size by percent of equity (2 percent risk per trade is a sane default). Because mean reversion has a high win rate but occasional large losers, a hard stop matters more than usual: place it well below entry (e.g. 2x ATR or a fixed percent) so a trade that keeps falling does not wreck the month. Never average down beyond a pre-set limit; adding to a loser is how mean reversion blows up.

WATCH OUT FOR

The high win rate lulls people into removing the stop, then one trade that does not revert gives back months of gains. It also stops working in bear markets, which is exactly when the dips look most tempting. The 200-day filter is load-bearing; do not drop it.

EVIDENCE

Connors and Alvarez documented RSI-2 extensively on US equity indices with strong historical win rates. It is one of the most-replicated short-term mean-reversion rules. Edge has compressed as it became well known, so treat modern expectations as more modest than the original backtests.

Copy into Claude Code — builds this strategy on your bot template

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Build a daily RSI-2 mean-reversion strategy as a class that plugs into my BaseStrategy interface. Logic: compute the 2-period RSI and the 200-day SMA. Generate a long signal (conviction toward +1) only when the close is above the 200-day SMA AND RSI(2) is below the entry threshold (default 10). Exit (conviction to 0) when the close rises back above the 5-day SMA or RSI(2) rises above 50, or after a max-hold of 10 trading days, whichever comes first. Expose rsi_period, rsi_entry, trend_ma, exit_ma, and max_hold as config parameters with the defaults above. Do not short. Return diagnostics: current RSI(2), distance above/below the 200-day SMA, and days in trade. Include a docstring explaining that this is a pullback-in-uptrend strategy that should be regime-gated to calm/bull regimes and must keep a hard stop. Write unit tests covering: signal fires only above the 200-day, exit on the 5-day cross, and the time stop. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Bollinger Band Reversion

Fade moves that pierce the outer band back toward the middle band.

HOW IT WORKS

Plot a moving average (the middle band, e.g. 20-day) with bands at plus/minus 2 standard deviations. Buy when price closes below the lower band (stretched cheap), exit when it returns to the middle band. Optionally short when it closes above the upper band in a range-bound market. The bands adapt to volatility, so signals self-scale as the market gets noisier or calmer.

WHERE THE EDGE COMES FROM

Statistical overextension. In a range-bound market, price spends most of its time near the mean, so a 2-sigma stretch is more likely to revert than to keep going.

WHEN TO USE IT

Best in range-bound, mean-reverting markets. As a standalone it whipsaws badly in trends, so gate it: only fade bands when a trend filter says the market is sideways, or only take the long side in an uptrend.

BEST REGIME

Range-bound, low-volatility

AVOID IN

Strong trend (price 'walks the band' and you fade a freight train), crash

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily bars for swing use; intraday bands work on liquid futures but get noisier the shorter the bar.

DAY / CALENDAR

Neutral. Band reversion is regime-driven, not calendar-driven.

ASSET CLASS

Index ETFs, large caps, liquid futures. Works on any liquid, mean-reverting instrument; weakest on persistently trending single names.

TIMEFRAME

Daily or intraday; holding 1 to 5 days on daily

ASSETS

Index ETFs, large caps, liquid futures

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
ma_period	20	Middle band lookahead.
num_std	2.0	Band width in standard deviations. 2.0 standard; 2.5 for fewer signals.
entry	close below lower band	Long entry. Short on close above upper band only in range regimes.
exit	touch of middle band	Or a fixed profit target; some exit at the opposite band.
trend_filter	ADX < 20 or flat 100-day MA	Only fade bands when the market is range-bound.

RISK & SIZING — SET TO YOUR OWN PLAN

Percent-of-equity sizing. Stop goes a defined distance beyond the band (e.g. 1 ATR below the lower band) because 'price walking the band' in a trend is the main failure mode and you need to be out fast. If shorting the upper band, size smaller; upside squeezes are violent.

WATCH OUT FOR

In a trend, price rides the outer band for a long time and naive fading produces a string of losses. The fix is a trend/range filter. Also, bands widen in volatility, so a 2-sigma event in a crash is not a reversion signal, it is a falling knife.

EVIDENCE

Band-reversion is widely used and documented; its edge is real in range regimes and negative in trends, which is why regime conditioning matters so much. Performance is regime-dependent, not universal.

Copy into Claude Code — builds this strategy on your bot template

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Build a Bollinger Band mean-reversion strategy as a BaseStrategy class. Compute a 20-period middle band and plus/minus 2-standard-deviation bands. Long signal when the close is below the lower band; exit when price touches or crosses the middle band. Add an optional short side (close above the upper band) that is only enabled when a range filter is true (default: ADX below 20). Expose ma_period, num_std, a use_shorts flag, and the trend-filter threshold as config. Return diagnostics: percent-b (where price sits within the bands), band width, and the trend-filter state. Docstring must warn that this strategy loses in trends and should be gated to range-bound regimes. Unit tests: long fires below lower band, exit at middle band, shorts only fire when the range filter passes. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Z-Score Reversion

Trade the standardized distance of price from its own moving average.

HOW IT WORKS

Compute a rolling mean and rolling standard deviation of price (or of the price/MA ratio). The z-score is (price minus mean) divided by standard deviation. Enter long when z drops below a negative threshold (e.g. -2), exit when z returns toward 0. This is the engine underneath most mean-reversion and pairs strategies, expressed directly. It generalizes cleanly to spreads for pairs trading.

WHERE THE EDGE COMES FROM

Same overextension logic as Bollinger, but expressed as a continuous, tunable distance rather than a fixed band, which makes conviction-scaling natural.

WHEN TO USE IT

Standalone on single names in range regimes, and as the core signal for pairs/spread trading. Conviction can scale with the magnitude of z, which fits a sizing-by-conviction system well.

BEST REGIME

Range-bound, low-volatility, mean-reverting

AVOID IN

Trending, crash

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Flexible: daily for swing, intraday for liquid instruments. The lookback should match the bar (e.g. 20 daily bars vs 20 5-minute bars).

DAY / CALENDAR

Neutral.

ASSET CLASS

Single equities, ETFs, and spreads/pairs. The same engine drives pairs trading, where it extends to two-asset spreads regardless of the underlying class.

TIMEFRAME

Daily or intraday

ASSETS

Single equities, ETFs, and (as a spread) pairs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
lookback	20 to 60	Window for the rolling mean and std. Shorter = more reactive, noisier.
entry_z	-2.0	Enter long when z falls below this.
exit_z	0.0 to -0.5	Exit as z reverts toward the mean.
max_z_stop	-3.5	If z keeps falling past this, the relationship may be broken; stop out.
scale_by_z	true	Optionally scale position size with z up to a cap.

RISK & SIZING — SET TO YOUR OWN PLAN

Percent-of-equity, optionally scaled by the z magnitude up to a hard cap so a -4 z does not produce an oversized position. The max-z stop is the key control: a z that blows through your stop level usually means the mean has shifted, not that reversion is imminent.

WATCH OUT FOR

Z-score assumes the mean is stable. In a regime shift the 'mean' moves and z stays extreme while you bleed. Short lookbacks make z jumpy and overtrade. Do not treat an extreme z as automatically more attractive; past a point it signals a broken relationship.

EVIDENCE

The standardized-distance approach is the textbook basis for statistical mean reversion and pairs trading. It works where the series is genuinely stationary and fails where it is not, which is why a stationarity or regime check belongs alongside it.

Copy into Claude Code — builds this strategy on your bot template

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Build a z-score mean-reversion strategy as a BaseStrategy class. Compute a rolling mean and rolling standard deviation of the close over a configurable lookback (default 20), then the z-score. Long when  $z < \text{entry\_z}$  (default -2.0); scale conviction with  $|z|$  up to a cap if scale_by_z is true. Exit when  $z \geq \text{exit\_z}$  (default 0). Hard stop if  $z < \text{max\_z\_stop}$  (default -3.5), flat the position and log a possible regime break. Expose lookback, entry_z, exit_z, max_z_stop, and scale_by_z. Return diagnostics: current z, rolling mean, rolling std. Docstring should note this is the same engine used for pairs spreads and that it assumes a stable mean, so it must be regime-gated. Unit tests: entry below threshold, conviction scaling capped, hard stop on extreme z. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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VWAP Reversion (Intraday)

Fade intraday stretches away from the volume-weighted average price.

HOW IT WORKS

VWAP is the average price weighted by volume since the session open, and it acts as an intraday magnet because institutions benchmark fills to it. When price stretches a configurable distance above or below VWAP (often measured in standard-deviation bands around VWAP), fade back toward it. Exit on the touch of VWAP or a fixed target. Strictly an intraday, flat-by-close strategy.

WHERE THE EDGE COMES FROM

Institutional execution anchors to VWAP, creating real flow that pulls price back toward it intraday, especially in range-bound sessions.

WHEN TO USE IT

Intraday on liquid instruments. Best on range/balanced days; turn it off on strong trend days (gap-and-go, news) where price leaves VWAP and never looks back.

BEST REGIME

Balanced/range intraday sessions

AVOID IN

Trend days, news-driven directional sessions

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday only, and time-of-day matters a lot: skip the first 15 to 30 minutes (VWAP is unstable at the open) and flatten before the close. Midday balanced sessions are the sweet spot.

DAY / CALENDAR

Neutral on day-of-week, but avoid known low-liquidity sessions (half-days, the afternoon before a holiday) where VWAP behaves oddly.

ASSET CLASS

Liquid futures (ES, NQ) and high-volume ETFs and stocks. Needs real intraday volume for VWAP to mean anything; useless on thin names.

TIMEFRAME

Intraday (1 to 15 min bars); flat by close

ASSETS

Liquid futures (ES, NQ), high-volume ETFs and stocks

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>vwap_bands</code>	1.5 to 2.5 std	Distance from VWAP that triggers a fade.
<code>entry</code>	price beyond band	Long below lower band, short above upper band.
<code>exit</code>	touch of VWAP	Or a fixed R-multiple target.
<code>session_filter</code>	skip first 15-30 min	The open is too noisy; let VWAP stabilize.
<code>trend_day_filter</code>	skip if price trends away from VWAP	Detect and stand down on trend days.

RISK & SIZING — SET TO YOUR OWN PLAN

Smaller per-trade size than daily strategies because frequency and slippage are higher. Hard intraday stop beyond the band. Always flat by the close. A trend-day detector (e.g. consecutive closes on one side of VWAP, or a strong opening drive) should disable new fades.

WATCH OUT FOR

Trend days are the killer: fading a market that is walking away from VWAP all day produces repeated losses. The open is noisy and VWAP is unstable in the first minutes. Transaction costs eat intraday mean reversion fast, so it needs liquid instruments and realistic cost modeling.

EVIDENCE

VWAP-anchoring is well established in execution and intraday literature; the reversion tendency is strongest in balanced sessions and absent on trend days. It is regime/day-type dependent, not a constant edge.

Copy into Claude Code — builds this strategy on your bot template

```
Build an intraday VWAP reversion strategy as a BaseStrategy class operating on intraday bars. Compute session VWAP and standard-deviation bands around it. Long when price is below the lower band, short when above the upper band; exit on a touch of VWAP or a fixed R target. Skip the first configurable minutes of the session (default 30). Include a trend-day filter that disables new entries when price has closed on the same side of VWAP for N consecutive bars (default 6) or shows a strong opening drive. Force flat at session close. Expose vwap_bands, session_skip_minutes, trend_day_n, and the R target. Return diagnostics: distance from VWAP in std, VWAP value, trend-day state. Docstring must stress flat-by-close, higher cost sensitivity, and that trend days disable the strategy. Unit tests: entries beyond bands, VWAP-touch exit, trend-day filter disabling entries, forced close at session end. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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MR-05

Gap Fade

Fade opening gaps that have a statistical tendency to fill.

HOW IT WORKS

When a liquid name opens a configurable percent away from the prior close (a gap up or down) without major news, take the opposite side expecting the gap to partially or fully fill during the session. Exit at the prior close (gap fill) or a partial target, with a stop beyond the open. Works best on gaps that are emotional/overnight-driven rather than fundamental.

WHERE THE EDGE COMES FROM

Overnight overreaction. Many gaps are driven by thin overnight liquidity and sentiment, and reverse once full liquidity returns at the open.

WHEN TO USE IT

Intraday, single names and index ETFs. Critical filter: skip gaps caused by earnings or hard news, which gap for a reason and keep going (that is PEAD territory, the opposite trade).

BEST REGIME

Calm/range markets, non-news gaps

AVOID IN

Earnings/news gaps, crash mornings, strong trend regimes

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday, anchored to the open. Enter in the first hour as the gap shows whether it will fill; flatten by close. The edge lives in the morning.

DAY / CALENDAR

Watch Mondays (weekend news creates more real-news gaps to exclude) and earnings-season clusters.

ASSET CLASS

Liquid large caps and index ETFs. Gaps on illiquid names are unpredictable; the strategy assumes deep opening liquidity.

TIMEFRAME

Intraday; same-day

ASSETS

Liquid large caps, index ETFs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
min_gap	0.5% to 2%	Minimum gap size to consider. Too small = noise.
max_gap	~4%	Above this, the gap is probably real news; skip.
news_filter	exclude earnings/halts	Mandatory. News gaps trend, they do not fill.
target	prior close (full fill)	Or partial fill (e.g. 50 percent of the gap).
stop	beyond the opening extreme	If the gap extends past the open, stand aside.

RISK & SIZING — SET TO YOUR OWN PLAN

Per-trade percent sizing, smaller for wider gaps. The news filter is a risk control, not a nicety: fading an earnings gap is fading PEAD and can be a large loss. Hard stop beyond the opening print. Flat by close.

WATCH OUT FOR

Earnings and news gaps are the classic trap, they keep going and the 'fade' compounds losses. Crash-morning gaps are not fade candidates. Without a news/earnings calendar filter this strategy is dangerous.

EVIDENCE

Gap-fill tendencies are well documented intraday for non-news gaps; the effect inverts for earnings-driven gaps (PEAD), so the two strategies are mirror images selected by the news filter.

Copy into Claude Code — builds this strategy on your bot template

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Build a gap-fade strategy as a BaseStrategy class for intraday use. At the open, compute the gap as (open minus prior close)/prior close. Consider a fade only when the absolute gap is between min_gap (default 0.5 percent) and max_gap (default 4 percent) AND an earnings/news flag for the symbol is false (accept this flag as an input; do not trade the symbol if it is true or unknown for an earnings date). Short a gap up, long a gap down; target the prior close (configurable to a partial fill), stop beyond the opening extreme, flat by close. Expose min_gap, max_gap, fill_target_pct, and require an is_news/has_earnings input. Return diagnostics: gap percent, target, news-flag state. Docstring must state plainly that earnings/news gaps must be excluded because they trend (the inverse PEAD trade). Unit tests: no trade outside the gap band, no trade when news flag is true, correct fade direction, forced close. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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MR-06

Internal Bar Strength (IBS)

Buy when the close sits near the low of the day's range; a simple, robust ETF reversion.

HOW IT WORKS

$IBS = (\text{close} - \text{low}) / (\text{high} - \text{low})$, a value from 0 to 1 measuring where the close landed within the bar's range. A low IBS (close near the low) is short-term oversold. Buy when IBS is below a threshold (e.g. 0.2), exit the next day or when IBS is high (e.g. above 0.8). Dead simple, one input, and historically robust on equity index ETFs.

WHERE THE EDGE COMES FROM

Short-term overreaction captured by where the close sits in the range. A close on the lows after a down day often bounces.

WHEN TO USE IT

Standalone on index ETFs, especially mean-reverting ones. Combines well with a regime filter and with RSI-2 (they agree often but capture slightly different things).

BEST REGIME

Range-bound and bull ETFs, low-volatility

AVOID IN

Crash, strong downtrend

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily close signal; act next session. Not intraday.

DAY / CALENDAR

Neutral.

ASSET CLASS

Equity index ETFs specifically. IBS is an ETF effect; on single stocks the close-in-range signal is drowned out by idiosyncratic moves.

TIMEFRAME

Daily; holding 1 to 3 days

ASSETS

Equity index ETFs (SPY, QQQ, IWM, sector ETFs)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>ibs_entry</code>	0.2	Buy when IBS is below this (close near the low).
<code>ibs_exit</code>	0.8	Exit when IBS is above this (close near the high).
<code>max_hold</code>	3 days	Time stop.
<code>regime_gate</code>	low-vol/bull only	Optional but recommended.

RISK & SIZING — SET TO YOUR OWN PLAN

Percent-of-equity sizing with a hard stop (e.g. 2 ATR). Same discipline as other mean reversion: the win rate is high, so the temptation to drop the stop is high, and that is the failure mode.

WATCH OUT FOR

Works far better on broad ETFs than on single stocks (single-name news swamps the IBS signal). Like all reversion, it degrades in bear markets. Keep it on indices and gate it by regime.

EVIDENCE

IBS is a well-documented, much-replicated ETF mean-reversion factor with strong historical results on equity indices and a reputation for robustness because it has essentially one parameter. Edge is regime-dependent and has compressed somewhat with popularity.

Copy into Claude Code — builds this strategy on your bot template

```
Build an Internal Bar Strength (IBS) mean-reversion strategy as a BaseStrategy class for daily ETF data. Compute  $IBS = (close - low) / (high - low)$ , guarding against a zero range. Long when  $IBS < ibs\_entry$  (default 0.2); exit when  $IBS > ibs\_exit$  (default 0.8) or after  $max\_hold$  days (default 3). No shorts. Expose  $ibs\_entry$ ,  $ibs\_exit$ , and  $max\_hold$ . Return diagnostics: current IBS, days in trade. Docstring should note IBS works best on broad index ETFs, not single names, and should be regime-gated to calm/bull regimes with a hard stop. Unit tests: zero-range guard, long fires below entry, exit above exit threshold and on the time stop. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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MR-07

Overnight / Close-to-Open Reversion

Exploit the documented difference between overnight and intraday returns.

HOW IT WORKS

US equity index returns have historically been earned disproportionately overnight (close to next open) rather than intraday (open to close). The simplest version buys the close and sells the next open on broad ETFs. Variants condition on the prior day (e.g. only hold overnight after a down day, blending with reversion). It is a calendar/structural effect, not a chart pattern.

WHERE THE EDGE COMES FROM

Structural: a large share of the equity risk premium has historically accrued overnight, attributed to overnight news flow, risk-transfer, and futures-market dynamics.

WHEN TO USE IT

Standalone as an overlay on index ETFs, or blended with a reversion filter (hold overnight only after weak days). Low per-trade edge, so costs and execution matter.

BEST REGIME

Normal/bull equity regimes

AVOID IN

Crash regimes (overnight gaps can be large and adverse)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Defined entirely by time-of-day: enter at/near the close, exit at the next open. The whole strategy is a time-of-day trade.

DAY / CALENDAR

Day-of-week effects exist in overnight returns (e.g. weekend-adjacent sessions differ); test before conditioning on it.

ASSET CLASS

Equity index ETFs and futures. The overnight premium is an index-level phenomenon; do not expect it on single names or other asset classes.

TIMEFRAME

Daily, overnight holding period

ASSETS

Equity index ETFs and futures

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
instrument	broad index ETF/ future	Effect is strongest on indices, weaker/absent on single names.
conditioning	none or 'after down day'	Conditioning can improve the blend with reversion.
cost_model	realistic spread/ commission	Per-trade edge is small; include realistic costs.
regime_gate	risk-off the strategy in crash regimes	Overnight tail risk is real.

RISK & SIZING — SET TO YOUR OWN PLAN

Size modestly; the per-night edge is small and tail risk (a bad overnight gap) is the main hazard. A regime gate that reduces or removes overnight exposure in high-vol/crash regimes is the key control. Never lever this up chasing the small edge.

WATCH OUT FOR

Tiny edge per trade means transaction costs can erase it; this needs cheap execution and liquid instruments. Overnight gap risk is asymmetric in crashes. The effect has been studied heavily and may be partly arbitrated; size for a modest, possibly-decayed edge.

EVIDENCE

The overnight-return anomaly (close-to-open outperforming open-to-close on US indices) is extensively documented across long samples. It is real but small, regime-sensitive, and a candidate for decay, so it is best as an overlay rather than a standalone return engine.

Copy into Claude Code — builds this strategy on your bot template

```
Build a close-to-open overnight strategy as a BaseStrategy class for daily index-ETF data.
Base behavior: signal to hold the position overnight (enter near close, exit at next open).
Add an optional conditioning flag 'after_down_day' that only holds overnight when the prior
session closed down. Include a regime input and reduce or zero overnight exposure when the
regime is crash/high-vol. Require a realistic transaction-cost input and incorporate it into
expected edge in the docstring/diagnostics. Expose conditioning mode and the regime risk-off
behavior. Return diagnostics: prior-day return, conditioning state, regime state. Docstring
must state the per-night edge is small, costs are decisive, and overnight gap risk is the
main hazard. Unit tests: hold-overnight signal generation, conditioning logic, regime risk-
off zeroing exposure. Finally, do not treat the defaults in this prompt as final. Expose the
key parameters (sizing, stop, max loss per trade, and the main strategy settings) as
configurable inputs, and add a short comment block telling me to set sizing, risk, and the
regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that
this is a starting template to configure and backtest on my own data, not a plug-and-play
system.
```

Keltner Channel Reversion

Fade stretches outside ATR-based channels around an EMA, a volatility-aware cousin of Bollinger.

HOW IT WORKS

Keltner Channels place bands a multiple of ATR above and below an EMA (e.g. 20-EMA, 2 ATR). Because the bands use ATR rather than standard deviation, they react more smoothly to volatility than Bollinger Bands. Fade a close outside the band back toward the EMA in range conditions; some traders instead trade breakouts of the channel in trends, so the regime decides which way you use it.

WHERE THE EDGE COMES FROM

Volatility-scaled overextension. An ATR-defined stretch from the EMA is statistically far in calm conditions and tends to revert, while the ATR basis makes the bands less jumpy than standard-deviation bands during volatility clusters.

WHEN TO USE IT

Range-bound regimes for the reversion read; gate with a trend filter exactly like Bollinger. Pairs well as a confirmation alongside Bollinger or z-score since the ATR basis disagrees with them at useful moments.

BEST REGIME

Range-bound, low-to-moderate volatility

AVOID IN

Strong trend (price rides the channel), crash

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily for swing; intraday on liquid futures with the lookback matched to the bar. Not strongly time-of-day sensitive on daily.

DAY / CALENDAR

Neutral.

ASSET CLASS

Index ETFs, large caps, liquid futures. Any liquid instrument that ranges; weakest on persistently trending single names.

TIMEFRAME

Daily or intraday; holding 1 to 5 days on daily

ASSETS

Index ETFs, large caps, liquid futures

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
ema_period	20	Center line. EMA reacts faster than an SMA.
atr_period	10 to 20	ATR lookback for band width.
atr_mult	2.0	Band distance in ATRs. Wider = fewer signals.
mode	reversion in range / breakout in trend	Regime decides the direction of use.
trend_filter	ADX < 20 for reversion	Only fade in range conditions.

RISK & SIZING — SET TO YOUR OWN PLAN

Set sizing and stops to your own rules; the channel defines where price is stretched, not how much you should bet. A stop a defined distance beyond the band handles the trend-ride failure mode. If you flip to breakout mode in trends, the risk profile changes to trend-following (low win rate, trail winners), so do not mix the two reads on one position.

WATCH OUT FOR

In a trend, price walks the Keltner channel just as it walks Bollinger Bands, so unfiltered fading bleeds. Confusing the reversion use with the breakout use leads to taking signals in the wrong regime. ATR bands lag volatility changes slightly.

EVIDENCE

Keltner Channels are a long-established volatility-band tool; the reversion edge is regime-dependent like all band reversion, and the ATR basis is valued for smoother behavior than standard-deviation bands. Treat the edge as conditional on a range regime, not constant.

Copy into Claude Code — builds this strategy on your bot template

```
Build a Keltner Channel reversion strategy as a class on my BaseStrategy interface. Compute a center EMA (default 20) and bands at atr_mult (default 2.0) times ATR (default 14) above and below. In reversion mode (enabled when a range filter such as ADX below 20 is true), go long on a close below the lower band and exit toward the EMA; short on a close above the upper band in range conditions. Expose ema_period, atr_period, atr_mult, mode, and the range filter. IMPORTANT: do not hardcode my position size or stop. Expose them as parameters and add a clear comment that I must set sizing, stop distance, and max-loss-per-trade to my own risk rules before running, and that the defaults are placeholders. Return diagnostics: distance from EMA in ATRs, band values, range-filter state. Docstring should note this loses in trends and must be regime-gated, and that breakout mode is a different (trend) use that should not be mixed with reversion on one position. Unit tests: band math, long below lower band, range-filter gating, that sizing/stop are parameters not constants. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Trend & Momentum

Bet that what is moving keeps moving. Low win rate, fat right tail: a few big winners pay for many small losers. The hard part is sitting through the chop. A regime or volatility filter that keeps you out of range-bound markets is doing most of the work.

TR-01 Moving Average Crossover

TR-02 Donchian / Turtle Breakout

TR-03 Time-Series Momentum (Absolute)

TR-04 MACD Trend/Momentum

TR-05 ADX-Filtered Trend

TR-06 52-Week High Momentum

TR-07 Residual (Idiosyncratic) Momentum

Moving Average Crossover

The canonical trend filter: fast MA over slow MA is long, under is flat/short.

HOW IT WORKS

Two moving averages, a fast and a slow (e.g. 50 and 200, the 'golden/death cross', or faster pairs like 20/50). Long when the fast is above the slow, flat or short when below. It is the simplest trend-state classifier and is more useful as a regime/trend filter for other strategies than as a standalone money-maker.

WHERE THE EDGE COMES FROM

Trend persistence: markets that have been rising tend to keep rising over medium horizons. The crossover is a lagging but robust way to stay on the right side of a sustained move.

WHEN TO USE IT

As a trend filter on top of other strategies, or standalone on trending instruments. On choppy, range-bound assets it whipsaws; pair it with a volatility or ADX filter to suppress signals in ranges.

BEST REGIME

Sustained trend (bull or bear for the short side)

AVOID IN

Range-bound, choppy

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily to weekly bars. Time-of-day is irrelevant; this is a position-level trend state, not an entry-timing tool.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Indices, futures, trending large caps. The short side works better on futures and FX than on equity indices.

TIMEFRAME

Daily to weekly

ASSETS

Indices, futures, trending large caps

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
fast_ma	50 (or 20)	Faster average.
slow_ma	200 (or 50)	Slower average. 50/200 is the classic.
ma_type	SMA or EMA	EMA reacts faster, whipsaws more.
allow_shorts	false on equities indices	Short side works better on futures/FX.
chop_filter	ADX > 20 to enable	Suppress signals when no trend exists.

RISK & SIZING — SET TO YOUR OWN PLAN

Trend strategies have low win rates and rely on a few large winners, so do not cut winners early and do size stops to survive the many small losses. Percent-of-equity sizing; a volatility-scaled stop keeps risk steady as the trend's volatility changes. Expect long flat/whipsaw stretches.

WATCH OUT FOR

Whipsaw in ranges is the main cost; a chop/ADX filter helps a lot. The crossover lags, so you give back some of every trend at the turn. Faster MAs trade more and lose more to costs and whipsaw. It is a filter first, a strategy second.

EVIDENCE

Trend-following via moving averages is one of the most studied and longest-documented systematic approaches, robust across asset classes over long horizons, though with low win rates and significant drawdowns during trendless periods.

Copy into Claude Code — builds this strategy on your bot template

```
Build a moving-average crossover strategy as a BaseStrategy class. Compute a fast and slow moving average (defaults 50 and 200, configurable type SMA/EMA). Long (conviction +1) when fast > slow; flat when fast < slow; allow shorts only if allow_shorts is true. Add a chop filter that requires ADX above a threshold (default 20) for a new signal, suppressing entries in range-bound markets. Expose fast_ma, slow_ma, ma_type, allow_shorts, adx_threshold. Return diagnostics: the two MA values, their spread, ADX. Docstring should frame this as primarily a trend filter with a low win rate that needs patience and a chop filter. Unit tests: long/flat state transitions on cross, shorts gated by the flag, ADX filter suppressing entries below threshold. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```


Donchian / Turtle Breakout

Buy N-day highs, exit on M-day lows: the original Turtle trend system.

HOW IT WORKS

Track the highest high and lowest low over a lookback (e.g. 20 or 55 days, the Donchian channel). Go long on a breakout above the N-day high, exit on a break below a shorter M-day low (e.g. 10-day). The Turtles famously traded a 20-day entry / 10-day exit and a slower 55/20 system. Pure breakout trend following with a defined exit.

WHERE THE EDGE COMES FROM

Breakouts to new highs initiate or continue trends; capturing the start of sustained moves and riding them is the entire premise of managed-futures trend following.

WHEN TO USE IT

Standalone on diversified futures and trending instruments, ideally across many uncorrelated markets so the few big winners can emerge. Single-instrument equity use is choppy.

BEST REGIME

Trending, volatility-expansion

AVOID IN

Range-bound (false breakouts), low-volatility chop

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily bars. Breakout level is checked on the close or intraday against the daily channel; not a sub-daily strategy as written.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Futures are the classic home, across a diversified set. Trending ETFs and FX work; single equities are choppy and need diversification.

TIMEFRAME

Daily

ASSETS

Futures (the classic home), trending ETFs, FX

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
entry_lookback	20 or 55	New high over this many days triggers entry.
exit_lookback	10 or 20	Break of this low exits. Always shorter than entry.
atr_stop	2 ATR	Initial stop; the Turtles used N=ATR-based sizing and stops.
pyramiding	optional, capped	Turtles added to winners in ATR increments; cap it.
universe	diversified	Trend works best across many uncorrelated markets.

RISK & SIZING — SET TO YOUR OWN PLAN

Volatility-based position sizing is core to the Turtle method: size each position so a 1-ATR move is a fixed fraction of equity, which equalizes risk across instruments. Low win rate (often 30 to 40 percent), so the system depends on letting winners run far. Expect deep drawdowns in trendless years.

WATCH OUT FOR

False breakouts in range-bound markets bleed the account; diversification across markets is the historical answer. On a single equity it whipsaws. The hard part is psychological: sitting through many small losses and long flat periods waiting for the rare big trend.

EVIDENCE

The Turtle system is one of the most famous documented trend-following approaches, with public track records and extensive replication. Breakout trend following remains a core managed-futures strategy, robust over decades but with low win rates and large drawdowns.

Copy into Claude Code — builds this strategy on your bot template

```
Build a Donchian/Turtle breakout strategy as a BaseStrategy class. Compute the highest high over entry_lookback (default 20) and the lowest low over exit_lookback (default 10). Long when price breaks above the entry-lookback high; exit when price breaks below the exit-lookback low. Use ATR for an initial stop (default 2 ATR) and expose an optional capped pyramiding rule that adds in ATR increments up to a max number of units. Expose entry_lookback, exit_lookback, atr_stop, atr_period, and pyramiding settings. Return diagnostics: current Donchian high/low, ATR, units held. Docstring should explain ATR-based volatility sizing, the low win rate, the need for diversification across markets, and the expectation of long flat periods. Unit tests: entry on new high, exit on channel-low break, ATR stop placement, pyramiding cap. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

TR-03

Time-Series Momentum (Absolute)

If an asset's own past 12-month return is positive, be long; if negative, flat or short.

HOW IT WORKS

For each asset, look at its own trailing return over a lookback (classically 12 months, often skipping the most recent month). Positive past return: hold long. Negative: go flat or short. Unlike cross-sectional momentum, this compares each asset only to itself, so it can be fully invested, partly invested, or net short depending on how many assets are trending up.

WHERE THE EDGE COMES FROM

Absolute (time-series) momentum: an asset's own trend predicts its near-future return. Documented across equities, bonds, commodities, and currencies over long samples.

WHEN TO USE IT

Standalone across a diversified set of assets/futures, or as a tactical overlay (risk-on when your holdings are above trend, risk-off when below). It naturally de-risks in bear markets by going flat.

BEST REGIME

Trending markets across asset classes

AVOID IN

Sharp reversals/choppy transitions where the trend flips repeatedly

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly evaluation on the close. Time-of-day and intraday timing do not apply.

DAY / CALENDAR

Irrelevant; rebalance on a fixed calendar day (e.g. month-end).

ASSET CLASS

Cross-asset by design: equity indices, bonds, commodities, FX. The diversification across classes is where most of the benefit comes from.

TIMEFRAME

Monthly rebalance (daily check possible)

ASSETS

Cross-asset: equity indices, bonds, commodities, FX

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
lookback	12 months	Trailing return window. 3/6/12-month blends are common.
skip_recent	1 month	Skip the most recent month to avoid short-term reversal.
rebalance	monthly	How often to re-evaluate.
short_side	flat or short on negative	Flat is gentler; short adds return and risk.
vol_target	optional overlay	Scale exposure to a volatility target for steadier risk.

RISK & SIZING — SET TO YOUR OWN PLAN

Pairs naturally with volatility targeting: scale each position so its risk contribution is even, which smooths the equity curve. Going flat (rather than short) in downtrends is the conservative choice. Main risk is whipsaw at trend turns; a monthly (not daily) rebalance reduces churn.

WATCH OUT FOR

Trend reversals cause clustered losses as positions flip. Monthly rebalancing helps; reacting daily overtrades. The short side adds return historically but also adds crash-up risk. As with all trend, expect flat-to-down years when nothing trends.

EVIDENCE

Time-series momentum is one of the most robust documented anomalies (Moskowitz, Ooi, Pedersen and the broad managed-futures literature), persistent across asset classes and decades, and the backbone of trend-following funds. Returns are lower and choppier in recent years than in the long history.

Copy into Claude Code — builds this strategy on your bot template

```
Build an absolute (time-series) momentum strategy as a BaseStrategy class that can operate per-asset across a set of instruments. For each asset, compute the trailing return over lookback months (default 12), optionally skipping the most recent month (default skip 1). Signal long if the past return is positive, flat or short if negative per the short_side setting, rebalanced monthly. Add an optional volatility-target overlay that scales each position toward a target risk. Expose lookback, skip_recent, rebalance frequency, short_side, and vol_target. Return diagnostics: trailing return, sign, scaled weight. Docstring should distinguish this from cross-sectional momentum (each asset vs itself), note monthly rebalancing to avoid churn, and that flat-in-downtrend is the conservative default. Unit tests: signal sign matches trailing return, skip-month logic, monthly rebalance cadence, vol-target scaling. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

MACD Trend/Momentum

Trade the moving-average convergence/divergence line crossing its signal line.

HOW IT WORKS

MACD is the difference between a fast and slow EMA (classically 12 and 26), with a signal line that is an EMA of the MACD (classically 9). Long when MACD crosses above its signal line, exit/short when it crosses below. The histogram (MACD minus signal) shows momentum building or fading. It blends trend and momentum in one indicator.

WHERE THE EDGE COMES FROM

Captures shifts in medium-term momentum slightly before a raw price MA cross because it reacts to the rate of change of the moving-average spread.

WHEN TO USE IT

As a momentum confirmation layer or a standalone trend signal on trending instruments. Like all MA-based tools it whipsaws in ranges; a trend/volatility filter helps.

BEST REGIME

Trending, momentum-building

AVOID IN

Range-bound, choppy

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily bars typically; intraday is possible but whipsaws more. Not time-of-day sensitive.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Indices, large caps, futures, FX. A general-purpose momentum tool across liquid instruments.

TIMEFRAME

Daily (intraday possible)

ASSETS

Indices, large caps, futures, FX

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
fast_ema	12	Fast EMA.
slow_ema	26	Slow EMA.
signal_ema	9	Signal-line EMA.
use_histogram	optional	Trade histogram zero-cross or divergence instead of line cross.
trend_filter	price > 200-day for longs	Reduce counter-trend whipsaw.

RISK & SIZING — SET TO YOUR OWN PLAN

Percent-of-equity with a volatility-aware stop. Treat MACD as a momentum filter, not a precise timing tool: it lags and whipsaws, so a confirming trend filter and patience with the win rate matter. Avoid stacking many correlated MACD signals as if they were independent.

WATCH OUT FOR

MACD is a lagging, derivative indicator and generates frequent false crosses in sideways markets. 'MACD divergence' is popular but unreliable as a standalone signal. Best used to confirm a trend identified another way, not as the sole input.

EVIDENCE

MACD is among the most widely used momentum indicators; its crossover edge is modest and regime-dependent, strongest as a confirmation layer within a trend-following framework rather than as a standalone system.

Copy into Claude Code — builds this strategy on your bot template

```
Build a MACD trend/momentum strategy as a BaseStrategy class. Compute MACD (fast EMA default 12, slow EMA default 26) and its signal line (EMA default 9), plus the histogram. Long when MACD crosses above the signal line; exit or short when it crosses below. Add an optional 200-day trend filter that only allows longs above the 200-day MA, and an option to trade the histogram zero-cross instead of the line cross. Expose fast_ema, slow_ema, signal_ema, use_histogram, and the trend filter. Return diagnostics: MACD, signal, histogram, trend-filter state. Docstring should state MACD lags, whipsaws in ranges, and works best as confirmation with a trend filter. Unit tests: line-cross long/short, histogram-mode toggle, trend filter gating longs. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

ADX-Filtered Trend

Only take trend signals when ADX confirms a trend actually exists.

HOW IT WORKS

ADX (Average Directional Index) measures trend strength regardless of direction, on a 0 to 100 scale. The strategy is a wrapper: take your directional trend signal (MA cross, breakout, etc.) only when ADX is above a threshold (e.g. 25), meaning a real trend is present, and stand aside when ADX is low (range-bound). Direction comes from the +DI/-DI lines or the underlying trend rule.

WHERE THE EDGE COMES FROM

Not a standalone edge but a regime conditioner: it removes the single biggest cost of trend strategies, which is whipsaw during trendless periods.

WHEN TO USE IT

As a filter layered on any trend or breakout strategy. This is arguably the most useful 'meta' tool in the trend family because trendless chop is where trend systems bleed.

BEST REGIME

Confirms when to deploy trend logic

AVOID IN

N/A (it is the filter that defines avoid)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Matches whatever strategy it wraps. As a filter it has no independent timing.

DAY / CALENDAR

Inherits the underlying strategy.

ASSET CLASS

Any trend-traded instrument; it is asset-agnostic because it only measures trend strength.

TIMEFRAME

Matches the underlying trend strategy

ASSETS

Any trend-traded instrument

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>adx_period</code>	14	ADX lookback.
<code>adx_threshold</code>	25 (20 looser, 30 stricter)	Above this, trend is 'on'.
<code>direction_source</code>	+DI/-DI or underlying rule	How to pick long vs short once trend is confirmed.
<code>exit_on_adx_fall</code>	optional	Exit when ADX drops back below threshold (trend fading).

RISK & SIZING — SET TO YOUR OWN PLAN

Inherits the sizing/stop rules of whatever trend strategy it wraps. Its job is to reduce the number of bad-regime trades, which improves the underlying strategy's risk-adjusted return rather than changing per-trade sizing.

WATCH OUT FOR

ADX lags and rises after a trend is underway, so it can keep you out of the very start of a move. A too-high threshold means few trades; too-low defeats the purpose. It tells you trend strength, not direction, so you still need a directional rule.

EVIDENCE

ADX as a trend-strength filter is well established; combining it with directional trend rules to suppress range-bound trades is a standard and effective technique in systematic trend following.

Copy into Claude Code — builds this strategy on your bot template

Build an ADX trend filter as a reusable BaseStrategy mixin/wrapper. Compute ADX and the +DI/-DI lines over `adx_period` (default 14). Expose a method that returns whether a trend is 'on' (ADX above `adx_threshold`, default 25) and the implied direction from +DI vs -DI. The wrapper should gate an underlying directional signal: pass it through only when trend is on, otherwise force flat. Optionally exit when ADX falls back below the threshold. Expose `adx_period`, `adx_threshold`, `direction_source`, and `exit_on_adx_fall`. Return diagnostics: ADX, +DI, -DI, trend-on state. Docstring should frame this as a regime conditioner that removes trendless-chop trades, note that ADX lags the start of trends, and that it gives strength not direction. Unit tests: trend-on threshold logic, direction from DI lines, gating an underlying signal to flat when off. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

TR-06

52-Week High Momentum

Buy stocks trading near their 52-week high; nearness to the high predicts continued gains.

HOW IT WORKS

Rank stocks by how close they trade to their trailing 52-week high (current price divided by the 52-week high). Names nearest their high have historically outperformed those far below, because traders anchor to the prior high and under-react when a stock pushes through it. Hold the names closest to their highs, rebalanced periodically. It is a cleaner, more robust cousin of standard price momentum.

WHERE THE EDGE COMES FROM

Anchoring and under-reaction at the 52-week high (George and Hwang): investors treat the prior high as a ceiling and are slow to re-rate a stock that breaks it, so proximity to the high predicts further gains better than raw past return in some studies.

WHEN TO USE IT

Cross-sectional, on a stock universe, monthly. Standalone momentum sleeve or a confirmation filter on other long signals. Regime-gate it: like all momentum it suffers crashes during sharp reversals.

BEST REGIME

Trending, dispersed markets

AVOID IN

Momentum-crash conditions (violent reversals out of winners), broad bear markets

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close. No intraday component.

DAY / CALENDAR

Irrelevant; rebalance on a fixed day.

ASSET CLASS

Liquid large- and mid-cap equities. An equity cross-sectional effect; needs a universe, not one ticker.

TIMEFRAME

Monthly rebalance

ASSETS

Liquid large- and mid-cap stock universe

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
proximity	price / 52-week high	The ranking metric; closer to 1.0 is stronger.
top_n	top decile or top N	How many of the nearest-high names to hold.
rebalance	monthly	Standard momentum cadence.
abs_filter	require uptrend / positive market	Avoid holding near-high names in a falling market.
min_price_liquidity	filter penny/ illiquid names	Keep the universe tradeable.

RISK & SIZING — SET TO YOUR OWN PLAN

Apply your own position sizing and concentration caps; equal-weight or volatility-weight the holdings and cap single-name weight. The momentum-crash tail is the main hazard, so an absolute-trend or market filter that moves to cash when the whole market rolls over is the key protection. Size for an edge that has compressed with publication.

WATCH OUT FOR

Shares momentum's crash risk: sharp reversals punish near-high portfolios hard. Without a liquidity filter the screen surfaces untradeable names. The effect is well known and partly arbitrated, so expect less than the original research magnitude.

EVIDENCE

The 52-week-high momentum effect (George and Hwang, 2004) is a well-documented anomaly where nearness to the 52-week high predicts returns and in their work explained much of standard momentum; it is robust across samples but, like all momentum, crash-prone and decayed since publication.

Copy into Claude Code — builds this strategy on your bot template

```
Build a 52-week-high momentum strategy as a class on my BaseStrategy interface, operating over a stock universe. Each monthly rebalance, compute proximity = current price / trailing 52-week high for each name, rank descending, and hold the top decile (or top N), with a liquidity/price filter to drop untradeable names. Add an absolute filter that moves to cash when the broad market is below its own trend. Expose proximity metric, top_n, rebalance frequency, the absolute filter, and liquidity thresholds. IMPORTANT: expose position sizing, weighting, and concentration caps as parameters with a comment that I must set them to my own risk rules; do not hardcode them. Return diagnostics: proximity ranking, holdings, cash weight. Docstring should note the anchoring/under-reaction basis, the momentum-crash tail, and the need for the absolute filter and a liquidity screen. Unit tests: proximity ranking, top-N selection, liquidity filter, absolute filter to cash, sizing as parameters. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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TR-07

Residual (Idiosyncratic) Momentum

Rank momentum on the part of returns not explained by market and factor exposure; higher Sharpe, smaller crashes.

HOW IT WORKS

Standard momentum mixes a stock's own trend with its factor exposures (market, size, value), which is part of why it crashes. Residual momentum first regresses each stock's returns on factor returns over a trailing window, then ranks names by the momentum of the residual (the stock-specific part). Hold high-residual-momentum names, short or avoid low ones. Stripping out factor exposure historically roughly doubles the risk-adjusted return and cuts the crash risk versus plain momentum.

WHERE THE EDGE COMES FROM

Firm-specific under-reaction isolated from factor noise. Conventional momentum's crashes come largely from time-varying factor (especially market) exposure; removing that leaves a cleaner, more persistent stock-specific momentum signal.

WHEN TO USE IT

Advanced cross-sectional equity sleeve for members comfortable running a factor regression. A better-behaved replacement for plain cross-sectional momentum. Still regime-aware, but less crash-prone than total-return momentum.

BEST REGIME

Trending, dispersed equity markets

AVOID IN

Still vulnerable to sharp reversals, though less than plain momentum

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close. No intraday element.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Liquid equities with available factor data. An equity cross-sectional strategy; the residualization needs factor returns.

TIMEFRAME

Monthly rebalance

ASSETS

Liquid equity universe (factor data required)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>factor_model</code>	CAPM (market) or Fama-French 3	Market-only already captures most of the benefit.
<code>estimation_window</code>	36 months	Window for the residual regression.
<code>momentum_lookback</code>	11 months, skip 1	Momentum of the residuals, classic 12-1 style.
<code>top_n</code>	top decile / quintile	Holdings from the residual ranking.
<code>rebalance</code>	monthly	Standard.

RISK & SIZING — SET TO YOUR OWN PLAN

Use your own sizing and concentration limits; volatility-weighting fits this well. Residual momentum is less crash-prone than plain momentum but not immune, so keep a market/regime filter. The extra moving part is the regression, so guard against thin estimation windows and look-ahead in the factor data.

WATCH OUT FOR

More complex to build and validate than plain momentum (you need clean factor returns and a careful, look-ahead-free regression). Some research argues part of the benefit is a betting-against-beta effect rather than pure stock-specific momentum, so do not oversell the mechanism. Decayed since publication like all factors.

EVIDENCE

Residual/idiosyncratic momentum (Blitz, Huij and Martens, 2011, and follow-ups) is well documented to deliver roughly double the risk-adjusted profit of conventional momentum with materially lower crash risk, and is used by quantitative managers such as Robeco and AQR; most of the gain comes from removing market-factor exposure.

Copy into Claude Code — builds this strategy on your bot template

```
Build a residual (idiosyncratic) momentum strategy as a class on my BaseStrategy interface over an equity universe, assuming factor return series are available. For each stock, regress its returns on the chosen factor model (default market-only/CAPM; optional Fama-French 3) over an estimation window (default 36 months), take the residuals, then compute residual momentum as the residual return over the last 11 months skipping the most recent month. Rank and hold the top decile (mirror short optional). Critically, perform the regression and ranking with no look-ahead (only data available at each rebalance). Expose factor_model, estimation_window, momentum_lookback, top_n, and rebalance frequency. IMPORTANT: expose sizing, weighting, and caps as parameters with a comment that I set them to my own risk rules; do not hardcode. Return diagnostics: per-name residual-momentum score, ranking, holdings. Docstring should explain it strips factor exposure to cut crashes, note the no-look-ahead requirement, and that part of the benefit may be a betting-against-beta effect. Unit tests: residual regression correctness, no-look-ahead in estimation, ranking/selection, sizing as parameters. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Cross-Sectional / Rotation

Rank a basket of assets against each other and hold the leaders, rotate out of laggards. This is momentum applied across names rather than across time, and it is one of the more durable documented effects. Needs a universe, not a single ticker.

CS-01 Relative-Strength Rotation

CS-02 Dual Momentum

CS-03 Sector Momentum Rotation

CS-04 Low-Volatility Factor

CS-05 Accrual Anomaly

Relative-Strength Rotation

Rank a universe by trailing return and hold the top N performers.

HOW IT WORKS

Each rebalance (monthly), rank every asset in a universe by its trailing return (e.g. 6 or 12 months, often skipping the last month). Hold the top N, sell anything that drops out of the top N. Equal-weight or rank-weight the holdings. This is momentum applied across names rather than across time, and it is one of the most durable documented anomalies.

WHERE THE EDGE COMES FROM

Cross-sectional momentum: winners keep outperforming losers over 3 to 12-month horizons, attributed to under-reaction and slow information diffusion. Documented since Jegadeesh and Titman (1993).

WHEN TO USE IT

Standalone on a sector-ETF or large-cap universe, or as the equity sleeve of a broader system. Needs a universe, not a single ticker. Pairs well with an absolute-momentum overlay (only hold names that are also above their own trend) to avoid being long the 'best of a falling market'.

BEST REGIME

Trending, dispersed markets

AVOID IN

Sharp reversals and 'momentum crashes' (violent rotations out of recent winners)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close. No intraday component.

DAY / CALENDAR

Irrelevant; pick a consistent rebalance day.

ASSET CLASS

Baskets: sector ETFs, country ETFs, large-cap universes. Needs a universe of comparable assets, not a single ticker.

TIMEFRAME

Monthly rebalance

ASSETS

Sector ETFs, country ETFs, large-cap baskets

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
lookback	6 to 12 months	Ranking window.
skip_recent	1 month	Avoid short-term reversal contaminating the rank.
top_n	top 3 to 5 (or top quintile)	How many to hold.
rebalance	monthly	Weekly overtrades; quarterly is slower.
abs_filter	require positive own-trend	Optional: only hold names also above their own trend.

RISK & SIZING — SET TO YOUR OWN PLAN

Equal-weight or volatility-weight holdings; cap single-name weight. The signature risk is the 'momentum crash', a sudden violent reversal where recent winners get dumped, so an absolute-momentum filter (go to cash when the whole universe is below trend) and volatility targeting are the main protections.

WATCH OUT FOR

Momentum crashes after market bottoms are brutal and clustered. Pure cross-sectional momentum can be fully long the best of a bear market; the absolute-momentum filter fixes this. High turnover if rebalanced too often. Crowding has compressed the premium.

EVIDENCE

Cross-sectional momentum is among the most robust and replicated anomalies across markets and asset classes, with decades of evidence, but it is prone to infrequent severe crashes and its premium has narrowed as it became widely harvested.

Copy into Claude Code — builds this strategy on your bot template

Build a relative-strength rotation strategy as a BaseStrategy-style class that operates over a universe of symbols. Each monthly rebalance: compute each symbol's trailing return over lookback months (default 6), skipping the most recent month (default 1), rank them, and target holding the top_n (default 4), equal-weighted. Add an optional absolute-momentum filter that drops any holding whose own trailing return is negative (go partially to cash instead). Expose lookback, skip_recent, top_n, rebalance frequency, and the abs_filter flag. Return diagnostics: the ranked table, current holdings, and cash weight. Docstring should distinguish cross-sectional from time-series momentum, warn about momentum crashes, and recommend the absolute-momentum filter plus volatility weighting. Unit tests: ranking and top-N selection, skip-month logic, absolute filter forcing cash on negative trends, monthly cadence. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Dual Momentum

Combine relative and absolute momentum: pick the strongest asset, but only if it beats cash.

HOW IT WORKS

Gary Antonacci's framework. First, relative momentum: compare assets (e.g. US stocks vs international stocks) and pick the stronger. Second, absolute momentum: only hold it if its trailing return also beats T-bills (cash); otherwise hold bonds/cash. So you own the strongest risk asset in up-trends and step to safety when even the leader is below cash.

WHERE THE EDGE COMES FROM

Stacks two documented effects: relative momentum (own the winner) and absolute momentum (avoid bear markets by comparing to cash). The cash comparison is what cuts the deep drawdowns.

WHEN TO USE IT

Standalone as a simple, robust tactical allocation between a few asset buckets. Excellent for members who want a low-maintenance, monthly, mechanical system rather than active trading.

BEST REGIME

Persistent trends; defensive automatically in bears

AVOID IN

Whipsaw transitions where it flips between risk and safety

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Asset-class ETFs (US equity, intl equity, bonds, plus a cash/T-bill proxy). Built for cross-asset allocation, not single-name trading.

TIMEFRAME

Monthly rebalance

ASSETS

Asset-class ETFs (e.g. US equity, intl equity, bonds, cash/T-bills)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
assets	2+ risk assets + a safe asset	E.g. SPY, EFA, with AGG/BIL as the safe haven.
lookback	12 months	Antonacci's canonical window.
abs_benchmark	T-bill / cash return	The absolute-momentum hurdle.
rebalance	monthly	Standard.
hold_n	1 (winner-take-all) or more	Classic dual momentum holds the single winner.

RISK & SIZING — SET TO YOUR OWN PLAN

Low-turnover and self-defensive: the absolute filter moves you to bonds/cash in downtrends, which historically cut equity-like drawdowns substantially. Main control is honoring the monthly cadence (no mid-month tinkering) and a sane asset menu.

WATCH OUT FOR

Whipsaw months where it flips risk-on/off near the threshold cost a little. Winner-take-all concentration means it can sit entirely in one asset. Look-back sensitivity: results vary with the chosen window, so do not over-fit it.

EVIDENCE

Dual momentum is a well-documented, widely replicated tactical framework (Antonacci) combining two robust effects, historically delivering equity-like returns with materially smaller drawdowns. Like all momentum, recent-decade performance is more muted than the long backtest.

Copy into Claude Code — builds this strategy on your bot template

```
Build a dual-momentum strategy as a BaseStrategy-style class over a small asset menu (configurable list of risk assets plus one safe asset). Monthly: compute each risk asset's trailing 12-month return (configurable). Relative step: pick the risk asset with the highest trailing return. Absolute step: hold it only if its trailing return exceeds the safe asset's (or a T-bill proxy's) return; otherwise hold the safe asset. Support hold_n (default 1, winner-take-all). Expose the asset menu, lookback, the safe asset, rebalance frequency, and hold_n. Return diagnostics: trailing returns of all assets, the chosen holding, risk-on/off state. Docstring should explain the two stacked filters and that the absolute step is what cuts drawdowns. Unit tests: relative selection, absolute filter moving to safe asset, monthly cadence. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Sector Momentum Rotation

Rotate among the eleven equity sectors, holding the strongest.

HOW IT WORKS

A focused relative-strength rotation on the sector ETFs (technology, financials, energy, etc.). Each month, rank sectors by trailing return and hold the top few. Because sectors lead and lag the cycle, rotation captures the leadership rather than holding the whole market through laggards.

WHERE THE EDGE COMES FROM

Sector dispersion and rotation through the business cycle: leadership persists month to month, and concentrating in leaders beats the cap-weighted average over time.

WHEN TO USE IT

Standalone equity sleeve, or as a more granular alternative to broad-index momentum. Pairs with an absolute filter (cash when no sector is above its own trend) and with regime gating.

BEST REGIME

Trending markets with clear sector leadership

AVOID IN

Correlated sell-offs where all sectors fall together (rotation has nowhere to hide)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

The 11 US sector ETFs. Equity-only by construction; the cash filter is the only non-equity element.

TIMEFRAME

Monthly rebalance

ASSETS

The 11 US sector ETFs (XLK, XLF, XLE, XLV, etc.)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
universe	11 sector ETFs	The standard sector set.
lookback	3 to 12 months	Shorter is more reactive.
top_n	top 2 to 3 sectors	Concentration vs diversification.
abs_filter	cash if sector below own trend	Avoid holding the best of a falling market.
rebalance	monthly	Standard.

RISK & SIZING — SET TO YOUR OWN PLAN

Equal-weight the held sectors and cap concentration. In broad sell-offs sectors correlate to ~1, so the absolute/cash filter is the real protection. Volatility weighting can steady the ride.

WATCH OUT FOR

In risk-off events sector correlations spike and rotation stops diversifying; only the cash filter helps. Too-short lookbacks chase noise and overtrade. Sector definitions and ETF overlaps (e.g. tech vs communication services) can blur the signal.

EVIDENCE

Sector rotation on relative strength is a well-documented application of cross-sectional momentum with solid historical support, sharing momentum's general robustness and its vulnerability to correlated drawdowns.

Copy into Claude Code — builds this strategy on your bot template

```
Build a sector-momentum rotation strategy as a BaseStrategy-style class over the 11 US sector ETFs (configurable list). Monthly: rank sectors by trailing return over lookback months (default 6) and hold the top_n (default 3), equal-weighted. Add an absolute-momentum filter that moves a holding's weight to cash if that sector's own trailing return is negative. Expose universe, lookback, top_n, abs_filter, and rebalance frequency. Return diagnostics: ranked sector table, holdings, cash weight. Docstring should note sector correlations spike in sell-offs so the cash filter is the key protection, and recommend volatility weighting. Unit tests: ranking/selection, absolute filter to cash, monthly cadence. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Low-Volatility Factor

Hold the lowest-volatility names; they have historically beaten high-vol names risk-adjusted.

HOW IT WORKS

Rank a universe by trailing realized volatility (or beta) and overweight the lowest-volatility quintile, underweight or avoid the highest. Rebalance periodically. The low-volatility anomaly is the counterintuitive finding that low-risk stocks have delivered better risk-adjusted (and sometimes absolute) returns than high-risk stocks, contradicting naive CAPM intuition.

WHERE THE EDGE COMES FROM

The low-volatility anomaly: leverage constraints and a lottery-preference for volatile stocks cause high-vol names to be overpriced and low-vol names underpriced, persistently.

WHEN TO USE IT

As a defensive equity sleeve or a tilt, not a fast trading strategy. Useful for members who want lower-drawdown equity exposure. Often combined with a quality or momentum factor.

BEST REGIME

Choppy and falling markets (defensive outperformance)

AVOID IN

Speculative bull runs where high-beta junk leads (low-vol lags)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly or quarterly rebalance on the close. A slow tilt, not a timing tool.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Large-cap equity universe, or a low-vol factor ETF as a shortcut. Equity-focused.

TIMEFRAME

Monthly or quarterly rebalance

ASSETS

Large-cap equity universe, or low-vol factor ETFs as a shortcut

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
vol_lookback	1 to 12 months	Window for realized vol/beta ranking.
hold_quintile	lowest 20%	Or a target number of lowest-vol names.
rebalance	quarterly	Low-turnover by nature.
weighting	inverse-vol or equal	Inverse-vol leans further into the effect.
combine_with	quality/momentum	Optional multi-factor blend.

RISK & SIZING — SET TO YOUR OWN PLAN

Low turnover, lower volatility by construction. The main 'risk' is relative: it underperforms in speculative melt-ups led by high-beta names, which tests patience. Size as a long-term tilt, not a trade; cap single-name weight.

WATCH OUT FOR

It can lag badly in frothy bull markets, leading people to abandon it right before it shines. Interest-rate sensitivity (low-vol stocks skew toward bond-like sectors) can hurt in rising-rate regimes. It is a slow factor, not a timing tool.

EVIDENCE

The low-volatility anomaly is extensively documented across global equity markets and decades, one of the better-supported factor findings, though it experiences long stretches of relative underperformance and has some sector/rate sensitivity.

Copy into Claude Code — builds this strategy on your bot template

```
Build a low-volatility factor strategy as a BaseStrategy-style class over an equity universe. Each rebalance (default quarterly): compute trailing realized volatility over vol_lookback (default 6 months) for each name, rank ascending, and hold the lowest quintile (or a target count), weighted inverse-vol or equal per config. Expose vol_lookback, hold fraction/count, rebalance frequency, and weighting. Return diagnostics: vol ranking table, holdings, weights. Docstring should explain the low-volatility anomaly, that it is a slow defensive tilt that lags in speculative bull markets, and note rate/sector sensitivity. Unit tests: vol ranking, lowest-quintile selection, inverse-vol weighting, rebalance cadence. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Accrual Anomaly

Favor companies whose earnings are backed by cash, avoid those propped up by accruals.

HOW IT WORKS

Sloan's accrual anomaly: firms with high accruals (earnings driven by accounting estimates rather than cash flow) tend to underperform, while low-accrual, cash-backed earners outperform, because the market over-weights reported earnings and under-weights their quality. Rank a universe by accruals (e.g. the accrual component of earnings scaled by assets), go long low-accrual names and underweight or short high-accrual ones. Rebalanced after earnings/financials update, typically annually or quarterly.

WHERE THE EDGE COMES FROM

Earnings-quality mispricing: investors fixate on headline earnings and fail to discount the lower persistence of the accrual component, so high-accrual firms are overpriced and low-accrual firms underpriced.

WHEN TO USE IT

A fundamental, low-turnover equity sleeve for members willing to pull accounting data. Combines naturally with value and quality factors. Not a trading strategy, a slow tilt rebalanced on fundamentals.

BEST REGIME

Broad equity markets over long horizons

AVOID IN

Speculative runs where low-quality, high-accrual names lead

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Quarterly/annual rebalance on the close, tied to financial-statement releases. No intraday element.

DAY / CALENDAR

Irrelevant; driven by the fundamentals calendar.

ASSET CLASS

Equities with fundamental data, excluding sectors where accruals are ill-defined. A fundamental equity tilt.

TIMEFRAME

Quarterly or annual rebalance

ASSETS

Equity universe with fundamental data

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
accrual_measure	balance-sheet or cash-flow accruals	Accruals scaled by average total assets.
rank	long low-accrual, short/avoid high-accrual	Top vs bottom decile.
rebalance	after financial-statement updates	Quarterly or annual.
combine_with	value / quality	Stronger as part of a multi-factor blend.
universe_filter		Accruals are ill-defined for some sectors.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
	<code>exclude</code> <code>financials / tiny</code> <code>caps</code>	

RISK & SIZING — SET TO YOUR OWN PLAN

Use your own sizing and caps; this is a diversified, low-turnover tilt, not a concentrated bet. The main exposure is style/relative: it lags when low-quality names lead. Exclude sectors where accruals are not meaningful (e.g. financials). Edge has compressed since publication.

WATCH OUT FOR

Requires accounting data and careful, point-in-time financials to avoid look-ahead (using restated or future-known figures inflates results). It can underperform for long stretches in junk-led rallies. The effect has weakened since it became widely known. Not meaningful for all sectors.

EVIDENCE

The accrual anomaly (Sloan, 1996) is one of the most studied accounting-based anomalies, with extensive evidence that low-accrual firms outperform high-accrual firms risk-adjusted; like other published anomalies it has attenuated over time and is best used as a diversified, point-in-time fundamental tilt.

Copy into Claude Code — builds this strategy on your bot template

Build an accrual-anomaly strategy as a class on my BaseStrategy interface over an equity universe with point-in-time fundamental data. Compute an accrual measure per firm (accruals scaled by average total assets), rank ascending, go long the lowest-accrual decile and optionally short/underweight the highest, excluding sectors where accruals are ill-defined (e.g. financials) and tiny/illiquid names. Rebalance after financial-statement updates (quarterly or annual). Critically, use only point-in-time data available at the rebalance date (no restated or future figures) to avoid look-ahead. Expose the accrual measure, ranking, rebalance cadence, universe filters, and a flag to combine with value/quality. IMPORTANT: expose sizing and caps as parameters with a comment that I set them to my own risk rules. Return diagnostics: accrual ranking, holdings, sector exclusions. Docstring should explain the earnings-quality basis, the point-in-time data requirement, that it lags in junk-led rallies, and that it has decayed. Unit tests: accrual computation, point-in-time data use, ranking/selection, sizing as parameters. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Volatility & Options

Trade volatility itself, or harvest the premium that option sellers collect on average. The volatility risk premium is real and persistent, but selling premium is picking up small steady gains in front of occasional large losses, so risk control is the entire game.

VOL-01 ATR Volatility Breakout

VOL-02 Volatility Targeting Overlay

VOL-03 VIX Term-Structure / Vol Risk Premium

VOL-04 Short-Premium Income (Spreads)

VOL-05 Covered Call / Buy-Write Overlay

ATR Volatility Breakout

Enter when price moves more than a multiple of recent ATR from a reference, riding the expansion.

HOW IT WORKS

Define a reference (prior close, session open, or a moving average) and enter when price moves beyond it by a multiple of ATR (e.g. open plus 0.5 ATR for a long). Volatility expansion often precedes or accompanies the start of a directional move; the ATR multiple makes the trigger self-scaling across instruments and regimes. Exit on a trailing ATR stop or end of day.

WHERE THE EDGE COMES FROM

Volatility clustering and breakout dynamics: a sharp expansion out of a quiet base tends to initiate a directional move as stops and breakout traders pile in.

WHEN TO USE IT

Intraday or daily on instruments prone to clean expansions (index futures, liquid stocks). Standalone, or as the entry trigger inside a trend system. Suppress in dead, low-range conditions where 'breakouts' are just noise.

BEST REGIME

Volatility-expansion, trend-initiation

AVOID IN

Low-volatility chop (false breakouts), illiquid names

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday or daily. Intraday versions are time-of-day sensitive: opening-range and mid-session breakouts behave differently, and you flatten by close.

DAY / CALENDAR

Neutral, though scheduled-news days (CPI, FOMC, jobs) produce more expansion; size for it.

ASSET CLASS

Index futures, liquid large caps, FX. Needs instruments that produce clean volatility expansions.

TIMEFRAME

Intraday or daily

ASSETS

Index futures (ES, NQ), liquid large caps, FX

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>atr_period</code>	14	ATR lookback.
<code>atr_mult</code>	0.5 to 1.0	Multiple of ATR beyond the reference to trigger.
<code>reference</code>	open or prior close or MA	What the breakout is measured from.
<code>trail_stop</code>	2 ATR	Trailing stop to ride the move.
<code>session</code>	intraday flat-by-close optional	If intraday, define the session and flat rule.

RISK & SIZING — SET TO YOUR OWN PLAN

Volatility-scaled sizing (size so 1 ATR is a fixed fraction of equity) keeps risk steady across instruments and over time. Breakouts have a low-ish win rate with good payoff, so use a trailing stop and let winners run. Beware clustering: several correlated breakouts at once is one bet, not many.

WATCH OUT FOR

False breakouts in quiet regimes are the main cost; an ATR or range filter that requires recent expansion helps. Gaps can blow through stops. Intraday versions are cost-sensitive. Do not shrink the ATR multiple to force more trades.

EVIDENCE

Volatility-breakout systems (e.g. ATR/range expansion entries) are a long-standing, documented family in futures and intraday trading, effective in expansion regimes and poor in quiet chop, hence the need for a volatility/regime filter.

Copy into Claude Code — builds this strategy on your bot template

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Build an ATR volatility-breakout strategy as a BaseStrategy class. Compute ATR over atr_period (default 14) and a reference level (default session open). Long when price exceeds reference + atr_mult*ATR (default 0.5), short when below reference - atr_mult*ATR. Use a trailing ATR stop (default 2 ATR) to ride the move; if intraday, force flat at session close. Add a filter requiring recent volatility expansion (current ATR above its own moving average) to suppress quiet-chop breakouts. Expose atr_period, atr_mult, reference, trail_stop, and the expansion filter. Return diagnostics: ATR, reference, breakout distance, expansion-filter state. Docstring should explain ATR-based sizing, the low win rate / high payoff profile, and false breakouts in quiet regimes. Unit tests: long/short trigger math, trailing stop, expansion filter suppression, flat-by-close. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Volatility Targeting Overlay

Not a signal: scale any strategy's exposure inversely to recent volatility for steadier risk.

HOW IT WORKS

Measure recent realized volatility of the strategy or asset and scale position size to hit a constant target volatility (e.g. 10 percent annualized). When markets get volatile, you hold less; when calm, more. This overlay smooths the equity curve and, because volatility clusters and spikes accompany drawdowns, it tends to de-risk into trouble automatically.

WHERE THE EDGE COMES FROM

Volatility is persistent and forecastable (it clusters), and high-volatility periods tend to have worse risk-adjusted returns, so targeting constant volatility improves Sharpe and trims tail losses.

WHEN TO USE IT

As an overlay on almost any strategy, especially trend and momentum. This is one of the highest-value 'meta' techniques in the whole library; it changes how much you hold, not what you trade.

BEST REGIME

Improves most strategies; especially valuable around volatility spikes

AVOID IN

Can lag a sudden vol spike (realized vol is backward-looking)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Matches the underlying strategy; rescaling is typically daily or weekly on the close.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Asset-agnostic. It scales size on any strategy in any class.

TIMEFRAME

Matches the underlying strategy; vol estimated over 20 to 60 days

ASSETS

Any

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
target_vol	10% annualized	The risk level you want to run at.
vol_lookback	20 to 60 days	Window to estimate realized volatility.
max_leverage	cap (e.g. 1.5x)	Hard cap so low-vol periods do not over-lever.
floor	minimum exposure	Avoid going to ~0 in extreme vol if undesired.
rebalance	daily or weekly	How often to rescale.

RISK & SIZING — SET TO YOUR OWN PLAN

This IS a risk tool. The hard leverage cap is essential so that quiet markets do not lever you into a latent blow-up. Realized vol is backward-looking, so it under-reacts to the first day of a spike; some implementations blend in an implied-vol or fast estimate. Keep the cap conservative.

WATCH OUT FOR

Backward-looking volatility lags the onset of a crash, so you de-risk a bit late. Over-frequent rescaling adds turnover and cost. Without a leverage cap, a long calm stretch can quietly build dangerous leverage. It reduces, not eliminates, tail risk.

EVIDENCE

Volatility targeting is extensively documented to improve risk-adjusted returns and reduce drawdowns for trend and equity strategies; it is standard practice in managed futures and risk-managed equity products. Benefits are clearest where volatility is persistent.

Copy into Claude Code — builds this strategy on your bot template

```
Build a volatility-targeting overlay as a reusable component that wraps any strategy's raw target position. Estimate realized volatility over vol_lookback (default 20 days, annualized) and compute a scaling factor = target_vol / realized_vol, clamped between a floor and max_leverage (default cap 1.5x). Apply the factor to the underlying strategy's position. Expose target_vol, vol_lookback, max_leverage, floor, and rebalance frequency. Return diagnostics: realized vol, scaling factor, capped weight. Docstring must state this is a risk overlay (changes size, not signal), that the leverage cap is mandatory, and that realized vol lags the first day of a spike. Unit tests: scaling math, clamp at cap and floor, application to an underlying position. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

VIX Term-Structure / Vol Risk Premium

Harvest the volatility risk premium by trading VIX futures term structure (contango vs backwardation).

HOW IT WORKS

VIX futures are usually in contango (longer-dated higher than spot) because investors overpay for protection, so short-vol positions earn a roll yield over time. When the curve inverts to backwardation (a fear spike), the edge flips and short-vol gets crushed. The strategy is long-vol-product short / short-vol-product long depending on the curve state, gated hard by the term-structure sign. This is sophisticated and dangerous; size tiny.

WHERE THE EDGE COMES FROM

The volatility risk premium: implied volatility persistently exceeds realized volatility because investors pay up for insurance, so sellers of that insurance earn a premium on average, paid for by occasional severe losses.

WHEN TO USE IT

Advanced only, as a small sleeve. Must be regime-gated by the term-structure state (only harvest in contango, flatten or flip in backwardation). Never a core holding; the tail is fat and left-skewed.

BEST REGIME

Calm markets in contango

AVOID IN

Volatility spikes / backwardation (short-vol gets destroyed: see Feb 2018)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily, evaluated on the close against the VIX futures curve. Not intraday.

DAY / CALENDAR

Irrelevant, but the curve can shift hard around macro events; the circuit breaker handles those.

ASSET CLASS

VIX futures and VIX ETPs specifically. This is a volatility-product strategy and does not generalize to other assets.

TIMEFRAME

Daily

ASSETS

VIX futures, VIX ETPs (with extreme caution re: structure and decay)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>ts_signal</code>	front/second VIX future ratio	Contango if front < second; backwardation if inverted.
<code>entry</code>	short vol only in contango	Flatten or go long vol in backwardation.
<code>position_cap</code>	very small	This sleeve must be a small fraction of the book.
<code>circuit_breaker</code>	hard stop on vol spike	Pre-defined exit when VIX or the curve jumps.
<code>decay_awareness</code>	model ETP roll decay	If using ETPs, account for their structural decay.

RISK & SIZING — SET TO YOUR OWN PLAN

Position sizing here is dominated by tail risk, not by normal volatility: short-vol can lose multiples of a normal day in a single session (Feb 2018 'Volmageddon' wiped out short-vol ETPs). Hard caps, a term-structure gate, and a pre-set circuit breaker are mandatory. This is the most dangerous strategy in the library; many should skip it.

WATCH OUT FOR

Catastrophic left tail: years of small gains can vanish in one spike. VIX ETPs have structural decay and have gone to zero. Backwardation flips the edge instantly. This strategy has bankrupted funds; treat the warnings as literal.

EVIDENCE

The volatility risk premium and VIX term-structure roll are well documented and genuinely persistent, but the strategy's return profile is steady gains punctuated by rare catastrophic losses, and several short-vol products have blown up, so it demands strict risk control and tiny sizing.

Copy into Claude Code — builds this strategy on your bot template

```
Build a VIX term-structure / volatility-risk-premium strategy as a BaseStrategy class, with safety as the priority. Use a term-structure signal from the ratio of the front to second VIX future: contango (front < second) enables a short-vol position; backwardation (front > second) forces flat or a long-vol hedge. Enforce a very small hard position cap and a pre-set circuit breaker that flattens on a configurable VIX spike or curve inversion. If trading ETPs, include a note/parameter accounting for roll decay. Expose the term-structure thresholds, position_cap (small default), and circuit-breaker level. Return diagnostics: term-structure ratio, regime (contango/backwardation), distance to circuit breaker. Docstring must state in plain terms that this strategy has a catastrophic left tail (reference the 2018 short-vol blow-up), must be tiny, and must be gated by term structure with a hard circuit breaker. Unit tests: contango enabling short-vol, backwardation forcing flat/hedge, circuit-breaker flattening, position cap enforcement. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Short-Premium Income (Spreads)

Sell defined-risk option spreads to collect premium in calm regimes.

HOW IT WORKS

Sell out-of-the-money options as defined-risk spreads (credit spreads, iron condors) to collect premium that decays in your favor (theta). Strikes are chosen near a target delta (e.g. 0.20 to 0.30), with a defined max loss from the long wing. Manage by closing at a profit target (e.g. 50 percent of max credit) or rolling/closing before expiration. Income comes from time decay plus the volatility risk premium.

WHERE THE EDGE COMES FROM

The volatility risk premium plus theta decay: implied vol usually exceeds realized vol, so systematically selling priced-in fear earns a premium, offset by losses when realized vol exceeds the premium collected.

WHEN TO USE IT

In calm or range-bound regimes on liquid underlyings (SPY, QQQ, large caps with liquid options). Regime gating is essential: stop selling premium when volatility regime turns or a crash filter fires. Pairs directly with your options bot guides.

BEST REGIME

Calm, range-bound, normal-to-elevated IV that then falls

AVOID IN

Crash and vol-spike regimes (short premium gets run over)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Manage daily on the close. Entries and the profit-target/DTE checks happen end-of-day; not an intraday strategy.

DAY / CALENDAR

Some sellers avoid opening new risk right before weekends or major events (weekend gap risk on short premium); a personal call.

ASSET CLASS

Liquid optionable ETFs and large caps. Options-only; the underlying must have tight, liquid options.

TIMEFRAME

Daily management; 20 to 45 DTE typical

ASSETS

Liquid optionable ETFs and large caps

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
target_delta	0.20 to 0.30	Short-strike delta; lower = safer, less credit.
dte_entry	20 to 45 days	Days to expiration at entry.
spread_width	defined-risk wing	Width sets max loss; size off it.
profit_target	50% of max credit	Close winners early to cut tail exposure.
dte_floor	close/roll by ~7-14 DTE	Avoid gamma/assignment risk near expiry.

RISK & SIZING — SET TO YOUR OWN PLAN

Size by MAX LOSS of the structure as a percent of equity, never by premium collected (a tiny credit can hide a large defined loss). Cap total short-premium exposure and net portfolio Greeks (especially short vega and short gamma). The whole game

is risk control: take profits early, respect the DTE floor, and cut size when the volatility regime turns. This is the income engine in your options track and inherits its guardrails.

WATCH OUT FOR

The classic trap is a smooth equity curve that masks a large hidden short-vol tail: it looks great until a gap or vol spike causes a loss many times the typical gain. Selling premium into a crash is the canonical way to blow up. Defined-risk spreads (not naked) and regime gating are non-negotiable.

EVIDENCE

Systematic premium selling harvests the well-documented volatility risk premium and works on average in calm regimes, but its left-tail risk is severe and concentrated in volatility spikes, which is why defined risk, profit-taking, and regime conditioning are emphasized everywhere it is taught responsibly.

Copy into Claude Code — builds this strategy on your bot template

```
Build a short-premium income strategy as a BaseStrategy class for options, assuming a Greeks/IV engine and a chain data layer exist. Each cycle (only when the regime input is calm/range): select a defined-risk credit spread or iron condor with short strikes near target_delta (default 0.25) at dte_entry (default 30), with a fixed spread width. Manage open structures: close at profit_target (default 50 percent of max credit), and close or roll by the dte_floor (default 10 DTE). Size by the structure's MAX LOSS as a percent of equity (default 1 to 2 percent), and cap total short-premium exposure and net short vega/gamma. Halt new entries when the regime is crash/high-vol. Expose target_delta, dte_entry, spread_width, profit_target, dte_floor, and the max-loss percent. Return diagnostics: chosen strikes, net credit, max loss, net Greeks, regime state. Docstring must state: size by max loss not credit, defined-risk only, take profits early, respect the DTE floor, and never sell into a crash regime. Unit tests: strike selection by delta, max-loss sizing caps, profit-target and DTE-floor closes, regime gate halting entries. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Covered Call / Buy-Write Overlay

Hold stock and sell calls against it to harvest premium and the volatility risk premium on a long position.

HOW IT WORKS

Own (or buy) the underlying and systematically sell out-of-the-money call options against it (one call per 100 shares). You collect call premium each cycle, which adds income and modestly cushions downside, in exchange for capping upside above the strike. A buy-write does both legs at once. It is the most accessible way to harvest the volatility risk premium because the long stock makes the short call covered, not naked.

WHERE THE EDGE COMES FROM

The volatility risk premium plus call theta, captured on a long-equity base: implied vol usually exceeds realized, so selling covered calls earns premium on average, at the cost of giving up the right tail of returns.

WHEN TO USE IT

As an income/overlay on long equity holdings, especially in flat-to-moderately-up regimes. Good for members who already hold ETFs/stocks and want yield. Manage the strike and roll cadence; gate strike aggressiveness by regime and by your own view.

BEST REGIME

Flat to moderately rising, range-bound, elevated-IV

AVOID IN

Strong bull runs (your upside is capped) and sharp crashes (premium barely cushions a large drop)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Manage on the close; roll on a monthly or weekly cadence. Not intraday.

DAY / CALENDAR

Many roll on a consistent weekday or at standard expirations; some avoid opening new short calls right before major events. A personal call.

ASSET CLASS

Optionable ETFs and large caps you are willing to own. Equity-plus-options; the underlying must have liquid options.

TIMEFRAME

Roll monthly (or weekly); manage on the close

ASSETS

Optionable ETFs and large caps you are willing to hold

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
call_delta	0.20 to 0.30	Lower delta caps less upside but collects less premium.
dte	20 to 45 days	Roll cadence; shorter rolls collect more theta but need more management.
roll_rule	roll at profit target or near expiry	When to buy back and resell.
assignment_plan	accept or roll if ITM	Decide ahead what happens if the call goes in-the-money.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
coverage	1 call per 100 shares	Stay fully covered; never oversell calls.

RISK & SIZING — SET TO YOUR OWN PLAN

Size to your own equity-position rules; the call is covered by the shares, but your real risk is the long stock, so this does not remove downside, it only cushions it slightly. Decide the assignment plan in advance (accept being called away vs rolling) so a rallying underlying does not force an emotional decision. Never sell more calls than you have shares to cover.

WATCH OUT FOR

Caps upside in exactly the strong bull markets where you most want it, which is the classic regret. The premium is small relative to a real crash, so do not mistake it for downside protection. Selling too-close (high-delta) calls for more premium gives away too much upside. Over-managing weekly rolls adds cost and effort.

EVIDENCE

Covered-call / buy-write strategies are well documented (e.g. long-running buy-write indices) to harvest the volatility risk premium with lower volatility than buy-and-hold, while trailing in strong bull markets due to capped upside; the income is real but is not a substitute for downside protection.

Copy into Claude Code — builds this strategy on your bot template

```
Build a covered-call / buy-write overlay as a class on my BaseStrategy interface for options, assuming a chain data layer and a long underlying position. Each cycle, sell one out-of-the-money call per 100 shares held at a target delta (default 0.25) and DTE (default 30); manage by rolling at a profit target or near expiry, and follow a configurable assignment plan if the call goes in-the-money. Never sell more calls than shares held (always covered). Gate strike aggressiveness by a regime input. Expose call_delta, dte, roll_rule, assignment_plan, and coverage. IMPORTANT: expose position sizing for the underlying as a parameter with a comment that I set it to my own equity-risk rules, and state in the docstring that this cushions but does not remove downside. Return diagnostics: chosen strike, premium collected, net delta, coverage check, regime state. Docstring should note capped upside in strong bull markets, that premium is not crash protection, and the over-management cost of frequent rolls. Unit tests: one-call-per-100-shares coverage enforcement, strike selection by delta, roll and assignment logic, sizing as a parameter. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Event & Calendar

Trade around scheduled events and calendar effects: earnings, index rebalances, month-end, FOMC. Edges here come from predictable flows and slow information diffusion, not from chart shapes. Many have decayed as they got crowded, so size positions for a smaller, decayed edge.

EV-01 Post-Earnings Announcement Drift (PEAD)

EV-02 Earnings Volatility / IV Crush

EV-03 Turn-of-Month Effect

EV-04 FOMC Drift / Pre-Announcement Effect

EV-05 Index Rebalance / Reconstitution

EV-06 Overnight Earnings Avoidance / Drift Capture

EV-01

Post-Earnings Announcement Drift (PEAD)

After a big earnings surprise, the stock drifts in the surprise's direction for weeks.

HOW IT WORKS

Measure the earnings surprise, usually Standardized Unexpected Earnings (SUE), the surprise normalized by its own volatility, or the immediate price/overnight reaction. Go long the biggest positive surprises (and optionally short the biggest negative ones) shortly after the announcement and hold for a drift window of several weeks. The market under-reacts to earnings news and prices adjust gradually, not instantly.

WHERE THE EDGE COMES FROM

Under-reaction to earnings information: investors are slow to fully incorporate a surprise, so prices drift toward fair value over the following weeks. The 'granddaddy of underreaction anomalies'.

WHEN TO USE IT

As a multi-name, event-driven sleeve. Stronger in smaller, less-covered stocks (higher limits to arbitrage). Needs an earnings calendar and a surprise measure. It is the mirror image of gap-fading: here you trade WITH the post-earnings move, not against it.

BEST REGIME

Normal markets; effect is broad but has weakened over time

AVOID IN

Crash regimes (idiosyncratic drift swamped by market moves); mega-cap names where it is weakest

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Enter the day after the announcement (let the initial reaction settle), hold for weeks. The entry day is event-driven, not clock-driven.

DAY / CALENDAR

Driven by the earnings calendar, which clusters around specific weeks each quarter rather than days of the week.

ASSET CLASS

Broad equity universe, tilted to smaller and low-coverage names where the drift is strongest. Single-stock equity event strategy.

TIMEFRAME

Hold 1 to several weeks after earnings

ASSETS

Broad equity universe, tilted toward smaller / low-coverage names

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
surprise_metric	SUE (or earnings-reaction return)	How to rank surprises. SUE = surprise / std of past surprises.
entry_window	day after announcement	Enter after the initial reaction to align with the drift.
hold_period	5 to 60 days	Test multiple horizons; classic studies use weeks to a quarter.
universe_tilt	smaller / low-analyst-coverage	Where the drift is strongest.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>long_short</code>	long top SUE, optional short bottom	Long-only is simpler; long-short is more market-neutral.

RISK & SIZING — SET TO YOUR OWN PLAN

Diversify across many names (this is a portfolio effect, not a single-stock bet); per-name size small. Avoid concentration in one sector or one earnings date. The effect is statistical and has decayed, so size for a modest, possibly-shrinking edge and include realistic transaction costs. Beware the inverse trade (fading earnings gaps), which fights this drift.

WATCH OUT FOR

PEAD has weakened materially over the decades as SUE persistence declined and the anomaly got crowded; modern expectations should be lower than 1990s-era backtests. It is strongest where it is hardest to trade (small, illiquid, low-coverage names) and weakest in the liquid mega-caps most people trade. Mega-cap, single-name versions disappoint.

EVIDENCE

PEAD is one of the most extensively documented anomalies (Bernard and Thomas reported roughly 8-9 percent quarterly abnormal returns on extreme-SUE portfolios historically), but the drift has declined over time, is concentrated in low-coverage names, and net-of-cost edge for accessible large caps is far smaller today.

Copy into Claude Code — builds this strategy on your bot template

```
Build a Post-Earnings Announcement Drift (PEAD) strategy as a BaseStrategy-style class
operating over a universe with an earnings calendar input. For each stock that just reported,
compute a surprise rank using SUE (surprise divided by the standard deviation of past
surprises) or, if EPS data is unavailable, the earnings-day reaction return as a proxy. Enter
the day after the announcement: long the top surprise decile (optionally short the bottom
decile if long_short is enabled), hold for hold_period days (default 20), and size each
position small with broad diversification. Allow a universe tilt toward smaller/low-coverage
names. Expose surprise_metric, entry_window, hold_period, universe_tilt, and long_short.
Return diagnostics: per-name SUE/reaction, decile, days held. Docstring must state the effect
has weakened over time, is strongest in small/low-coverage names and weak in mega-caps, and
is the inverse of fading earnings gaps. Unit tests: surprise ranking, day-after entry timing,
hold-period exit, long-short selection. Finally, do not treat the defaults in this prompt as
final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy
settings) as configurable inputs, and add a short comment block telling me to set sizing,
risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note
clearly that this is a starting template to configure and backtest on my own data, not a
plug-and-play system.
```

Earnings Volatility / IV Crush

Sell the elevated option premium into earnings and profit from the post-report IV collapse.

HOW IT WORKS

Implied volatility ramps up into a scheduled earnings report (uncertainty), then collapses immediately after the news is out ('IV crush'). Defined-risk premium sellers (e.g. iron condors or short strangles converted to defined risk) sell that inflated premium before the report and close just after, capturing the IV drop. The bet is that the realized move is smaller than the priced-in expected move.

WHERE THE EDGE COMES FROM

Predictable IV inflation before earnings and collapse after: option buyers overpay for the event on average, so disciplined sellers of the event premium earn the difference when the actual move underperforms expectations.

WHEN TO USE IT

Advanced options sleeve on liquid optionable names with a clear earnings date. Requires understanding the expected move and accepting occasional large losses when a stock gaps far beyond the implied move. Defined risk only.

BEST REGIME

Names whose realized moves tend to undershoot the implied move

AVOID IN

High-flying names prone to huge earnings gaps (the implied move underestimates reality)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Strongly time-of-day defined: enter at the close the day before earnings (peak IV), exit at the open after (capture the crush). Do not hold mid-day.

DAY / CALENDAR

Follows each name's earnings date; cluster risk during peak earnings weeks.

ASSET CLASS

Liquid large/mega caps with active options and known earnings dates. Options-only.

TIMEFRAME

Open day before earnings, close day after

ASSETS

Liquid large/mega caps with active options and known earnings dates

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
structure	defined-risk (iron condor / spread)	Never undefined risk into a binary event.
short_delta	~0.15 to 0.20	Place shorts outside the expected move.
expected_move	from ATM straddle price	Set strikes relative to the priced-in move.
entry	close before the report	Capture peak IV.
exit		Capture the IV crush quickly; do not linger.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
	open after the report	

RISK & SIZING — SET TO YOUR OWN PLAN

Strictly defined-risk: earnings are binary events and undefined-risk sellers get destroyed by a large gap. Size by max loss per trade, small, and diversify across many earnings events rather than betting one. Expect a left-skewed distribution: many small wins, occasional max-loss when a stock gaps well beyond the expected move.

WATCH OUT FOR

A single large surprise gap can hit max loss instantly; the strategy lives or dies on diversification and defined risk. Picking individual names to sell into earnings is closer to gambling than the diversified version. Liquidity and wide spreads on the options can erode the edge.

EVIDENCE

The pre-earnings IV ramp and post-earnings IV crush are well documented and reliable in direction; the profitability of selling it depends on whether realized moves undershoot implied moves on average, which holds for many names but with a meaningful left tail, making defined risk and diversification essential.

Copy into Claude Code — builds this strategy on your bot template

```
Build an earnings IV-crush strategy as a BaseStrategy class for options, assuming a Greeks/IV engine, chain data, and an earnings calendar. For a name with earnings tomorrow: estimate the expected move from the ATM straddle price, then construct a DEFINED-RISK iron condor (or defined spread) with short strikes outside the expected move near short_delta (default 0.15-0.20). Enter at the close before the report, exit at the open after (capturing IV crush); do not hold longer. Size strictly by max loss as a small percent of equity and diversify across many earnings events. Expose structure, short_delta, expected_move source, and the max-loss percent. Return diagnostics: expected move, chosen strikes, net credit, max loss. Docstring must state earnings are binary, undefined risk is forbidden, size by max loss, diversify across events, and expect occasional max-loss gaps. Unit tests: expected-move calc, strike placement outside it, defined-risk construction, entry/exit timing, max-loss sizing. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Turn-of-Month Effect

Equity returns cluster around the last few and first few trading days of the month.

HOW IT WORKS

Historically, a disproportionate share of equity-index gains has occurred in a window spanning roughly the last one to three trading days of a month and the first two to three of the next, attributed to month-end fund flows, salary investing, and rebalancing. The simplest version holds a broad index ETF only during that window and sits in cash the rest of the month.

WHERE THE EDGE COMES FROM

Predictable calendar flows: pension contributions, payroll 401k investing, and institutional rebalancing concentrate buying around month-end and month-start.

WHEN TO USE IT

As a low-effort overlay or tilt on broad index exposure, or to time when a mean-reversion/other strategy is allowed to be long. Tiny standalone edge; better as a seasonal weight than a sole system.

BEST REGIME

Normal/bull equity regimes

AVOID IN

Crash regimes (flows do not save you in a sell-off)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily; enter near the close on the entry day, exit on the close of the exit day. The window is in calendar days, not clock time.

DAY / CALENDAR

Entirely day-of-month driven (last few and first few trading days). This is its defining feature.

ASSET CLASS

Broad equity-index ETFs and futures. An index-level flow effect.

TIMEFRAME

Daily; in-market a handful of days per month

ASSETS

Broad equity-index ETFs/futures

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
window_before	last 1 to 3 trading days	Days before month-end to enter.
window_after	first 2 to 3 trading days	Days after month-start to exit.
instrument	broad index ETF	Effect is an index-level flow phenomenon.
cost_model	realistic	Small edge; costs matter.
regime_gate	risk-off in crash regimes	Flows do not protect in a crash.

RISK & SIZING — SET TO YOUR OWN PLAN

Small size; the edge per occurrence is modest. Because you are only exposed a few days a month, standalone returns are low, so this is best as a tilt/overlay that decides when other long strategies may operate. A regime gate avoids being long into the window during a crash.

WATCH OUT FOR

It is a small, well-known, possibly-decaying effect; do not over-allocate to it. Exact window definition matters and is easy to over-fit. It will not protect you in a genuine sell-off that happens to straddle month-end.

EVIDENCE

The turn-of-month effect is well documented across decades and markets and is one of the more persistent calendar anomalies, but the per-trade magnitude is small and it is best treated as an overlay rather than a primary return source.

Copy into Claude Code — builds this strategy on your bot template

Build a turn-of-month overlay as a BaseStrategy class for a broad index ETF. Using a trading calendar, signal long during a window from window_before trading days before month-end (default 2) through window_after trading days after month-start (default 3), and flat otherwise. Include a regime input that forces flat during crash/high-vol regimes, and require a realistic cost model. Expose window_before, window_after, the instrument, and the regime gate. Return diagnostics: position of the current date in the month, in-window state, regime state. Docstring should note the per-trade edge is small, it is best as an overlay deciding when other longs may run, and it does not protect in crashes. Unit tests: window detection around month boundaries, flat outside the window, regime gate forcing flat. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

FOMC Drift / Pre-Announcement Effect

Equities have historically drifted up in the day before scheduled Fed announcements.

HOW IT WORKS

A documented 'pre-FOMC announcement drift' shows a large share of historical equity excess returns accruing in the ~24 hours before scheduled FOMC rate decisions. The simple version holds a broad index into the pre-announcement window and exits around the announcement. It is a narrow, scheduled-event effect tied to a handful of dates per year.

WHERE THE EDGE COMES FROM

A pre-announcement risk-premium/positioning effect: equities have historically risen ahead of scheduled FOMC meetings, a pattern documented by the New York Fed among others.

WHEN TO USE IT

As a small scheduled-event overlay (only around the eight FOMC dates per year), not a standalone system. Useful as an example of event-driven, calendar-anchored trading for the community to study.

BEST REGIME

Normal regimes

AVOID IN

When the meeting itself is the crisis (emergency/ contentious meetings behave differently)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Enter roughly a day before the scheduled announcement, exit at or just before the decision (usually early afternoon on FOMC days). Time-of-day around the announcement matters.

DAY / CALENDAR

Driven by the eight scheduled FOMC dates per year, not a weekly pattern.

ASSET CLASS

Broad equity-index ETFs and futures. Index-level.

TIMEFRAME

Hold ~1 day into the announcement, a few times per year

ASSETS

Broad equity-index ETFs/futures

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
event_calendar	scheduled FOMC dates	Drive entries off the known FOMC schedule.
entry	~1 day before announcement	Enter into the pre-announcement window.
exit	at/around the announcement	Exit before or right at the decision.
instrument	broad index	Index-level effect.
size	small	Few events per year; modest, possibly-decayed edge.

RISK & SIZING — SET TO YOUR OWN PLAN

Very small allocation; only a handful of events per year and the effect, while well documented historically, may attenuate as it becomes known. Exit discipline around the announcement matters because the post-announcement move is a coin flip. Do not lever a calendar curiosity.

WATCH OUT FOR

Only ~8 occurrences a year means tiny sample and high variance; one bad year tells you little. The effect is well publicized and may be decaying. Contentious or emergency meetings do not follow the pattern. Strictly an overlay/curiosity, not a core engine.

EVIDENCE

The pre-FOMC announcement drift is documented in research (notably a New York Fed study) showing outsized historical equity returns in the pre-announcement window, but it rests on few annual observations and may weaken as it is arbitrated, so it warrants only small, overlay-level use.

Copy into Claude Code — builds this strategy on your bot template

Build a pre-FOMC drift overlay as a BaseStrategy class for a broad index ETF, driven by a list of scheduled FOMC announcement dates (input). Signal long starting one trading day before each scheduled announcement and exit at or just before the announcement; flat otherwise. Keep size small and expose the event-date list, entry lead time, exit timing, and instrument. Return diagnostics: next FOMC date, in-window state. Docstring must note the small annual sample, possible decay, that emergency meetings differ, and that this is an overlay/curiosity not a core strategy. Unit tests: window detection around supplied FOMC dates, flat outside windows, correct exit timing. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Index Rebalance / Reconstitution

Trade the predictable flows when stocks are added to or removed from major indices.

HOW IT WORKS

When a stock is added to a major index (e.g. S&P 500), index funds must buy it, and removed names get sold, on a known schedule. Historically, additions tended to rise into the effective date (front-running the forced index-fund buying) and reverse somewhat afterward. The strategy positions ahead of the announced rebalance using the published methodology and effective dates.

WHERE THE EDGE COMES FROM

Predictable, mechanical demand: passive funds must trade index changes regardless of price, and the announcement-to-effective-date gap lets others position ahead of that forced flow.

WHEN TO USE IT

Event-driven, around scheduled reconstitution dates of major indices. Requires tracking index-change announcements and understanding each index's methodology. A niche but well-known flow-based edge.

BEST REGIME

Normal liquidity conditions

AVOID IN

Crowded trades where the effect is already arbitrated;
illiquid names with huge index-weight changes

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Position after the announcement and around the effective date; the timing is in days relative to those events, not intraday.

DAY / CALENDAR

Driven by each index's reconstitution schedule.

ASSET CLASS

Stocks being added to or removed from major indices. Single-name, event-driven equity.

TIMEFRAME

Days around announcement and effective date

ASSETS

Stocks being added to / removed from major indices

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>index_calendar</code>	reconstitution/ effective dates	Drive trades off the published schedule.
<code>entry</code>	after announcement, before effective date	Position ahead of forced index-fund flows.
<code>exit</code>	around/after the effective date	The flow completes; the edge often reverses after.
<code>liquidity_filter</code>	tradeable size vs index demand	Bigger relative index demand = bigger effect.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
crowding_awareness	edge has compressed	Many now trade this; size accordingly.

RISK & SIZING — SET TO YOUR OWN PLAN

Per-name size modest; the effect has compressed as it became well known and front-run. The reversal after the effective date is real, so exit discipline matters. Liquidity risk in the names with the largest weight changes can cut both ways.

WATCH OUT FOR

Crowding has shrunk the classic add/delete pop because so many traders anticipate it. The post-effective-date reversal can hurt latecomers. It requires accurate, timely index-change data and methodology knowledge. Edge is real but smaller than it was.

EVIDENCE

The index-inclusion effect (price impact from forced passive flows around reconstitution) is well documented historically, though academic and practitioner work shows it has weakened over time as anticipation and arbitrage increased, so it is now a smaller, more crowded edge.

Copy into Claude Code — builds this strategy on your bot template

```
Build an index-rebalance strategy as a BaseStrategy-style class driven by an input list of
index changes (symbol, action add/remove, announcement date, effective date). For additions,
go long after the announcement and exit around the effective date; for removals, the mirror
(short into the effective date) if shorting is enabled. Include a liquidity filter comparing
tradeable size to expected index demand and a note that the edge has compressed. Expose the
change-list input, entry/exit timing relative to dates, a shorting flag, and the liquidity
filter. Return diagnostics: days to effective date, action, estimated index demand. Docstring
must note crowding has shrunk the effect, the post-effective-date reversal, and the need for
accurate index-change data. Unit tests: entry/exit timing around supplied dates, add vs
remove direction, liquidity filter gating. Finally, do not treat the defaults in this prompt
as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy
settings) as configurable inputs, and add a short comment block telling me to set sizing,
risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note
clearly that this is a starting template to configure and backtest on my own data, not a
plug-and-play system.
```

Overnight Earnings Avoidance / Drift Capture

Systematically hold (or avoid) positions through earnings based on the post-earnings reaction, not a guess.

HOW IT WORKS

Two linked rules built on the same fact that earnings cause large, unpredictable overnight gaps. The avoidance version flattens or reduces any position in a name reporting after the close, removing binary gap risk from a non-earnings strategy. The capture version is the disciplined counterpart: only after the report, enter in the direction of the realized surprise (linking to PEAD) rather than betting on the announcement. Together they turn earnings from a random shock into a managed rule.

WHERE THE EDGE COMES FROM

Two effects: earnings gaps are large and unforecastable (so avoiding them removes uncompensated risk), and the post-earnings reaction carries information (so entering after, with the drift, is favorable). The edge is risk-control plus disciplined drift participation.

WHEN TO USE IT

As an overlay on any single-name equity strategy with an earnings calendar. The avoidance rule belongs on almost every single-stock system; the capture rule is for members who specifically want post-earnings exposure. Both require knowing each holding's earnings date.

BEST REGIME

Any (avoidance is universal); capture works in normal regimes

AVOID IN

Holding blind through earnings with no plan is what this exists to prevent

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Event-driven: act into the close before a report (avoidance) or after the open following it (capture). Time-of-day matters around the event.

DAY / CALENDAR

Driven by the earnings calendar, which clusters in specific weeks each quarter.

ASSET CLASS

Single-name equities with scheduled earnings. Not relevant to broad index or non-equity exposure.

TIMEFRAME

Event-driven around each holding's earnings date

ASSETS

Single-name equities with scheduled earnings

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
earnings_calendar	required input	Per-holding confirmed report dates.
pre_earnings_action	flatten / reduce / hold	What to do into a report (avoidance: flatten or reduce).
post_earnings_entry	optional, with the surprise	Capture mode: enter after, in the drift direction.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
size_through_event	reduced if held	If you do hold, size down for gap risk.
blackout_window	day of / day around report	Define the no-new-risk window.

RISK & SIZING — SET TO YOUR OWN PLAN

This is primarily a risk overlay, so the point is to apply your own reduced-size or flat rule into earnings rather than carry full size through a binary event. If using capture mode, size small and diversify across many earnings events, never concentrate on one name's report. Define the pre-earnings action and blackout window in advance so the rule is mechanical.

WATCH OUT FOR

The avoidance rule costs you the occasional favorable gap, which is the price of removing the unfavorable ones; do not abandon it after a missed pop. The capture rule is just PEAD and inherits its decay and small-cap tilt. Both fail if the earnings calendar is wrong or stale, so data quality is critical.

EVIDENCE

That scheduled earnings produce large, hard-to-predict overnight moves is well established, which is why systematic single-name strategies commonly impose earnings blackouts; the capture side rests on the documented (and decayed) post-earnings drift. The overlay is standard risk practice rather than a standalone return source.

Copy into Claude Code — builds this strategy on your bot template

```
Build an earnings overlay as a reusable component for my BaseStrategy single-name equity strategies, taking a per-holding earnings calendar as input. Implement two modes. Avoidance: in a configurable blackout window around a holding's report, flatten or reduce the position per pre_earnings_action so binary gap risk is removed; if held, size down. Capture: after the report, optionally enter in the direction of the realized surprise (a PEAD-style drift entry), small and diversified. Expose the earnings calendar input, pre_earnings_action, post_earnings_entry, size-through-event, and the blackout window. IMPORTANT: expose the reduced-size and any capture sizing as parameters with a comment that I set them to my own risk rules; do not hardcode. Return diagnostics: days to next earnings per holding, blackout state, action taken. Docstring should frame avoidance as universal single-name risk practice, capture as PEAD with its decay/small-cap tilt, and stress that a stale earnings calendar breaks the overlay. Unit tests: blackout detection around supplied dates, flatten/reduce logic, capture entry direction, sizing as parameters. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Pairs & Stat-Arb

Trade the spread between two related assets rather than direction. Market-neutral when it works. The danger is that the relationship breaks; a pair that was cointegrated stops being cointegrated and you hold the losing legs of both sides. Correlation-health monitoring is not optional.

PR-01 Static-Pairs Z-Score

PR-02 Cointegration Spread (Engle-Granger)

PR-03 Correlation-Regime Allocation

PR-04 ETF vs Constituents Arbitrage

Static-Pairs Z-Score

Trade the spread between two historically related assets back to its mean.

HOW IT WORKS

Pick two assets that move together (e.g. two stocks in the same industry, or an ETF and a close peer). Build the spread (a hedge-ratio-weighted difference of their prices), compute its z-score, and when the spread stretches ($|z| > 2$) go long the cheap leg and short the rich leg, expecting convergence. Exit when the spread reverts ($|z| < 0.5$). Market-neutral: you profit from the relationship, not market direction.

WHERE THE EDGE COMES FROM

Mean reversion of an economically linked spread: two similar assets should not diverge permanently, so temporary divergences tend to converge.

WHEN TO USE IT

Standalone market-neutral sleeve in low-volatility regimes. Use pre-selected, economically sensible pairs (same sector, or an ETF vs its near-substitute). It is the classic stat-arb starter, but it **MUST** be paired with correlation-health monitoring because the relationship can break.

BEST REGIME

Low-volatility, range-bound, stable correlations

AVOID IN

Crashes and regime shifts (correlations break, spreads diverge and do not return)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily on the close, or intraday for liquid pairs. Match the z-score lookback to the bar.

DAY / CALENDAR

Neutral.

ASSET CLASS

Sector-peer stocks and related ETFs, as predefined pairs. Equity/ETF spreads; the method generalizes to any cointegrated pair but start with equities.

TIMEFRAME

Daily (intraday possible)

ASSETS

Sector peer stocks, related ETFs (predefined pairs)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
pairs	predefined list	Economically linked pairs; do not data-mine random pairs.
hedge_ratio	OLS or rolling beta	Weights the legs so the spread is balanced.
entry_z	2.0	Enter when the spread's z exceeds this.
exit_z	0.5	Exit as the spread reverts toward the mean.
stop_z	3.5 to 4.0	If the spread keeps diverging, the relationship may be broken; stop.

RISK & SIZING — SET TO YOUR OWN PLAN

Size the pair as a single market-neutral unit (dollar-balanced legs). The stop on an extreme spread is essential: a spread blowing past your stop usually means the relationship broke, not that convergence is imminent. Never average into a diverging spread without a hard limit. Cap the number of simultaneous pairs to control correlation among the pairs themselves.

WATCH OUT FOR

Pairs decay: cointegration is not permanent, and the failure mode is holding the losing legs of both sides as the spread widens forever. This is one of the most reliable ways to lose money in quant if you skip correlation monitoring. Data-mined pairs that look cointegrated on small windows are usually spurious.

EVIDENCE

Pairs trading via spread mean reversion is a long-documented stat-arb approach (Gatev, Goetzmann, Rouwenhorst) with solid historical results that have decayed as the trade became crowded. Its profitability is conditional on stable relationships, which makes correlation-health monitoring the deciding factor.

Copy into Claude Code — builds this strategy on your bot template

```
Build a static-pairs z-score strategy as a BaseStrategy class (using my PairStrategy/multi-asset signal interface). Accept a list of predefined, economically linked pairs. For each pair: estimate a hedge ratio (rolling OLS beta), build the spread, and compute its rolling z-score. Enter when  $|z| > \text{entry\_z}$  (default 2.0): long the cheap leg, short the rich leg, dollar-balanced as one unit. Exit when  $|z| < \text{exit\_z}$  (default 0.5). Hard stop and flag a possible broken relationship if  $|z| > \text{stop\_z}$  (default 3.75). Cap the number of simultaneous pairs. Expose the pairs list, hedge-ratio method, entry_z, exit_z, stop_z, and max simultaneous pairs. Return diagnostics: spread, z-score, hedge ratio, legs. Docstring must warn that pairs decay, the failure mode is holding both losing legs, correlation-health monitoring is required, and data-mined pairs are usually spurious. Unit tests: hedge-ratio calc, entry/exit on z thresholds, dollar-balanced legs, hard stop on extreme z. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Cointegration Spread (Engle-Granger)

Only trade a pair after a statistical test confirms the spread is genuinely mean-reverting.

HOW IT WORKS

Before trading a pair, run a cointegration test (e.g. Engle-Granger: regress one leg on the other, then test the residual for stationarity with an ADF test). Only trade pairs that pass the test on a sufficient history. Then trade the cointegrating residual (the spread) on its z-score like a normal mean-reversion. Re-test periodically; drop pairs that lose cointegration. This adds statistical rigor to pairs selection.

WHERE THE EDGE COMES FROM

Same spread-reversion edge as static pairs, but with a statistical gate that filters out spurious relationships, improving the odds the spread actually reverts.

WHEN TO USE IT

For members who want to select pairs rigorously rather than by eye. Best on stable, economically linked candidates; the test confirms the linkage is statistically real over the sample. Still requires ongoing re-testing because cointegration is not permanent.

BEST REGIME

Stable regimes where tested relationships hold

AVOID IN

Regime shifts that break previously cointegrated relationships

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily on the close, with periodic (e.g. monthly) re-testing. Not intraday.

DAY / CALENDAR

Neutral; re-test on a fixed schedule.

ASSET CLASS

Candidate pairs of related instruments, usually equities/ETFs. Asset-agnostic in principle but most accessible in equities.

TIMEFRAME

Daily; periodic re-testing (e.g. monthly)

ASSETS

Candidate pairs of related instruments

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
candidate_pairs	economically sensible set	Test only plausible pairs; testing all pairs invites false positives.
test	Engle-Granger ADF on residual	Or Johansen for multivariate.
pvalue_threshold	0.05	Significance to accept cointegration.
min_history	1 to 2 years	Enough data for a meaningful test.
retest_frequency	monthly	Drop pairs that lose cointegration.

RISK & SIZING — SET TO YOUR OWN PLAN

Same single-unit market-neutral sizing and hard spread stop as static pairs. The added control is statistical: a multiple-testing problem if you scan too many pairs (some will pass by luck), so restrict candidates and treat marginal p-values skeptically. Re-test and retire pairs that decay.

WATCH OUT FOR

Testing many random pairs guarantees spurious 'cointegrated' pairs by chance (multiple-comparisons trap); restrict candidates to economically linked ones. Cointegration found in-sample can vanish out-of-sample. A passing test is necessary, not sufficient: the relationship can still break after you start trading it.

EVIDENCE

Cointegration-based pairs selection is the statistically principled version of pairs trading and is well established in the literature; it improves pair quality but does not eliminate relationship-breakdown risk, and is vulnerable to multiple-testing bias if applied indiscriminately.

Copy into Claude Code — builds this strategy on your bot template

```
Build a cointegration-spread strategy as a BaseStrategy class using my multi-asset interface. Accept a restricted set of economically sensible candidate pairs. For each, run an Engle-Granger test (regress leg A on leg B, ADF-test the residual) over min_history (default 1.5 years) and only enable pairs with p-value below pvalue_threshold (default 0.05). Trade the cointegrating residual on its z-score (entry |z|>2, exit |z|<0.5) as a dollar-balanced unit, with a hard spread stop. Re-test on retest_frequency (default monthly) and drop pairs that lose cointegration. Expose the candidate list, the test type, pvalue_threshold, min_history, and retest_frequency. Return diagnostics: test p-value, spread, z, hedge ratio. Docstring must warn about the multiple-testing trap (restrict candidates), that in-sample cointegration can vanish out-of-sample, and that a passing test does not prevent breakdown. Unit tests: cointegration test gating, residual spread construction, z-score entries/exits, retest dropping a decayed pair. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```

Correlation-Regime Allocation

Trade pairs only while their correlation is healthy; step away when it breaks down.

HOW IT WORKS

The complement to spread trading: monitor the rolling correlation (and/or the spread's behavior) of a pair, and scale pair exposure by correlation health. When correlation is strong and stable, trade the spread normally; when correlation weakens or breaks, cut or exit the pair before the spread runs away from you. This is the risk-management half that makes pairs trading survivable.

WHERE THE EDGE COMES FROM

Not a return signal but a survival filter: pairs strategies lose money precisely when relationships break, so detecting deterioration and stepping aside avoids the worst losses.

WHEN TO USE IT

As an overlay on any pairs/spread strategy. If you only ship spread-reversion without this, you ship the half that loses money when cointegration breaks. Essential, not optional, for a responsible pairs sleeve.

BEST REGIME

Keeps you trading while relationships hold

AVOID IN

N/A (it defines when to stop)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Matches the pairs strategy it overlays.

DAY / CALENDAR

Inherits the underlying.

ASSET CLASS

Any traded pairs; it is a risk overlay, not asset-specific.

TIMEFRAME

Matches the underlying pairs strategy

ASSETS

Any traded pairs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
corr_lookback	60 to 120 days	Rolling correlation window.
corr_floor	e.g. 0.6	Below this, reduce or exit the pair.
scale_mode	linear or on/off	Scale exposure with correlation, or hard-gate it.
spread_break_check	spread fails to revert in N days	Secondary breakdown detector.
exit_on_break	flatten the pair	When correlation breaks, get out, do not hope.

RISK & SIZING — SET TO YOUR OWN PLAN

This is a risk control; it reduces pair exposure as relationships weaken. Combined with the spread strategy's own stop, it provides two independent breakdown defenses. The key discipline is acting on deterioration early rather than waiting for the spread stop to be hit at a larger loss.

WATCH OUT FOR

Correlation is noisy on short windows, so too-short a lookback causes false breakdown signals and over-exiting. Correlation can stay high while the spread still misbehaves, so pair it with a spread-behavior check. The hard part is psychological: exiting a 'temporarily' broken pair instead of hoping it converges.

EVIDENCE

Conditioning pairs exposure on correlation/cointegration health is the standard professional response to pairs' central weakness (relationship breakdown) and materially reduces tail losses; it is widely recommended in stat-arb practice as the necessary complement to spread reversion.

Copy into Claude Code — builds this strategy on your bot template

Build a correlation-regime allocation overlay as a reusable component for pairs strategies, using my multi-asset interface. For each active pair, compute rolling correlation over `corr_lookback` (default 90 days) and scale the pair's exposure by correlation health: full exposure above `corr_floor` (default 0.6), reduced (linear) or zero (on/off) below it per `scale_mode`. Add a secondary breakdown detector that flags when the spread fails to revert within `N` days, and flatten the pair on a confirmed break. Expose `corr_lookback`, `corr_floor`, `scale_mode`, the spread-break window, and `exit_on_break`. Return diagnostics: rolling correlation, health scale, breakdown flags. Docstring must frame this as the survival half of pairs trading, warn that short windows give noisy correlation, and stress exiting broken pairs rather than hoping. Unit tests: correlation scaling, floor gating, spread-break detection, flatten-on-break. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

ETF vs Constituents Arbitrage

Trade the small gap between an ETF and a basket replicating its holdings.

HOW IT WORKS

An ETF should track the value of its underlying basket (its NAV). When the ETF price diverges from a replicating basket (or a close proxy), trade the spread expecting convergence: long the cheaper side, short the richer. For most liquid ETFs the gap is tiny and arbitrated by authorized participants, so the realistic community version targets less-efficient ETFs or uses a proxy basket and treats it as a spread-reversion trade with extra care on costs.

WHERE THE EDGE COMES FROM

Arbitrage between an ETF and its underlying value: persistent divergences should not exist for efficient ETFs, so gaps in less-efficient ones tend to converge.

WHEN TO USE IT

Advanced, and mostly illustrative for liquid US ETFs (the pros keep them efficient). More viable on less-liquid or harder-to-arbitrage ETFs, or as a teaching example of the ETF/NAV relationship. Costs and borrow can dominate the small edge.

BEST REGIME

Normal liquidity; less-efficient ETFs

AVOID IN

Stressed markets where ETF/NAV gaps blow out for real liquidity reasons (not a free arb)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday to daily. Liquid-ETF gaps close fast, so the realistic version is intraday with fast execution.

DAY / CALENDAR

Neutral, but stressed-market days widen gaps for real reasons; the regime filter handles those.

ASSET CLASS

ETFs versus their replicating baskets or proxies. ETF-specific.

TIMEFRAME

Intraday to daily

ASSETS

ETFs vs replicating baskets or proxies

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
etf_and_basket	ETF + replicating holdings/proxy	Full replication is hard; a proxy basket is common.
gap_threshold	cost-aware minimum	Only trade gaps bigger than round-trip costs.
entry	fade the ETF/NAV gap	Long cheap side, short rich side.
exit	gap convergence	Back to fair value.
cost_borrow_model	explicit	Costs and short borrow can exceed the edge.

RISK & SIZING — SET TO YOUR OWN PLAN

Tiny edge, so cost and borrow modeling are decisive; trade only gaps larger than realistic round-trip costs. In stressed markets ETF/NAV gaps widen for genuine liquidity reasons and are NOT a free arbitrage, so a regime/liquidity filter and a stop are needed. Size small; this is closer to a teaching example than a core engine for liquid ETFs.

WATCH OUT FOR

For liquid ETFs the gap is arbitrated away and net-of-cost edge is essentially zero. During stress, gaps widen because the underlying is hard to trade, not because of free money, and fading them can mean large losses. Full basket replication is operationally hard; proxies introduce tracking error.

EVIDENCE

ETF-versus-NAV arbitrage is a real mechanism that keeps liquid ETFs efficient (via authorized participants), which is exactly why retail-accessible edge is minimal in liquid names; documented opportunities concentrate in less-liquid ETFs and stressed conditions, where the 'arbitrage' carries real risk.

Copy into Claude Code — builds this strategy on your bot template

```
Build an ETF-versus-constituents spread strategy as a BaseStrategy class using my multi-asset interface, framed as advanced and illustrative. Accept an ETF and a replicating basket (or proxy). Compute the gap between the ETF price and the basket value, and only trade gaps larger than a cost-aware threshold: long the cheaper side, short the richer, exit on convergence. Require explicit transaction-cost and short-borrow models that feed the threshold, and a liquidity/regime filter that stands down in stressed conditions. Expose the ETF/basket definition, gap_threshold, and the cost/borrow models. Return diagnostics: ETF price, basket value, gap, cost-adjusted edge. Docstring must state that liquid ETFs offer near-zero net edge, stressed-market gaps are not free arbitrage, and proxy baskets add tracking error. Unit tests: gap calc, cost-threshold gating, direction of the fade, regime stand-down. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
```


Alternative Charting

Reframe price using bricks, ranges, or averaged candles to strip out noise. These are lenses, not edges by themselves: the signal still has to be a real trend or reversion. Renko and range bars in particular have backtesting traps that fool people constantly.

AC-01 Renko Trend

AC-02 Heikin-Ashi Trend Smoothing

AC-03 Point-and-Figure Breakout

AC-04 Range Bars

Renko Trend

Use fixed-size price bricks to strip out noise and trade the cleaned trend.

HOW IT WORKS

Renko charts plot a brick only when price moves a fixed amount (the brick size), ignoring time. A new brick in the same direction extends the trend; a reversal requires price to move two brick sizes the other way. Trade in the direction of the brick color, ideally with a confirmation filter (e.g. a moving average on the bricks, or waiting for two same-color bricks). The chart's job is to make trends visually clean by filtering small wiggles.

WHERE THE EDGE COMES FROM

Noise reduction: by requiring a minimum price move per brick, Renko filters the chop that whipsaws trend signals on time-based charts, leaving clearer trend structure.

WHEN TO USE IT

Trend-following on instruments with sustained moves, on daily-to-weekly horizons where the inherent lag is acceptable. NOT for scalping (the lag means the move is over by the time a brick prints). Confirm with a moving average to avoid range-bound whipsaw.

BEST REGIME

Sustained trends; higher timeframes

AVOID IN

Range-bound markets (bricks alternate and whipsaw), intraday/scalping

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily-to-weekly perspective. Explicitly NOT intraday or scalping; the brick lag means short timeframes underperform.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Trending FX pairs, indices, commodities. Trend-oriented instruments; poor on choppy, range-bound names.

TIMEFRAME

Daily to weekly perspective (longer holds)

ASSETS

Trending FX pairs, indices, commodities

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
brick_method	ATR-based	ATR brick adapts to volatility; fixed size is rigid. Use ~0.5-1.0x ATR(14).
brick_size	derived from ATR or fixed	LOCK THIS before backtesting; changing it repaints history.
confirmation	2-brick or MA filter	Avoid acting on a single false brick flip.
stop	1 to 2 bricks against entry	Where the trend structure breaks.
backtest_on	UNDERLYING time-based prices	Critical: do not backtest on synthetic brick prices.

RISK & SIZING — SET TO YOUR OWN PLAN

Volatility-aware sizing via the ATR brick. The non-obvious but critical risk control is methodological: Renko bricks are synthetic prices, so backtesting fills at brick prices is inaccurate, and changing brick size repaints the whole history. You must lock the brick size first and backtest against real time-based prices, or your results are fiction.

WATCH OUT FOR

Two big traps. First, REPAINTING: changing brick size redraws history, so a setup you 'see' today may not have existed, and naive backtests on brick prices are unreliable, set and lock the brick size before testing and test on underlying prices. Second, RANGE WHIPSAW: in sideways markets bricks alternate colors and treating every flip as a signal bleeds; a two-brick or MA filter is required. Also, Renko always lags, so you give back part of every trend at the turn and miss the early entry.

EVIDENCE

Renko is a widely used trend-clarity tool; practitioner backtests report meaningful whipsaw reduction in trends on higher timeframes, but charting platforms explicitly warn that Renko prices are synthetic and unsuitable for naive backtesting, and that brick-size changes repaint history. It is a lens that helps trend systems, not a standalone edge, and it is poor intraday.

Copy into Claude Code — builds this strategy on your bot template

```
Build a Renko trend strategy as a BaseStrategy class, with correctness safeguards. Construct Renko bricks from the UNDERLYING time-based price series using an ATR-based brick size (default ~0.5-1.0x ATR(14)), computed once and then LOCKED so history does not repaint. Generate the brick series and trade in the brick direction, requiring a confirmation filter (default: two consecutive same-color bricks AND price on the correct side of a moving average applied to bricks). Stop one to two bricks against entry. Critically, for any backtest, fill orders at real time-based prices at the moment the brick would confirm, NOT at synthetic brick prices. Expose brick_method, ATR multiple, confirmation mode, and stop distance. Return diagnostics: current brick direction, brick size, confirmation state. Docstring MUST warn: brick size must be locked before testing (repainting), backtest on underlying prices not brick prices, range-bound whipsaw requires the confirmation filter, and Renko lags so it is unsuitable for scalping. Unit tests: brick construction from prices, ATR brick sizing, two-brick confirmation, that backtest fills use underlying prices. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Heikin-Ashi Trend Smoothing

Averaged candles that smooth price into cleaner trend runs for easier holding.

HOW IT WORKS

Heikin-Ashi ('average bar') candles recompute open/close from averages of the current and prior bar, producing smoother candles where strong trends show long same-color runs with little opposite-side wick. Trade in the candle direction; consider exiting when candles change color or show indecision (long wicks both sides). Unlike Renko, Heikin-Ashi keeps the time axis, so it is easier to backtest cleanly.

WHERE THE EDGE COMES FROM

Visual smoothing that helps a trader stay in a trend by muting the bar-to-bar noise that triggers premature exits; the edge is behavioral/holding-discipline more than a new signal.

WHEN TO USE IT

As a trend-holding aid or a confirmation layer on top of a trend strategy, on daily timeframes. It helps you ride trends without being shaken out, but it lags and the smoothed prices are not tradeable prices.

BEST REGIME

Trending markets

AVOID IN

Range-bound (frequent color flips), and any need for precise entry/exit prices

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily bars. A trend-holding aid, not an intraday timing tool.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Indices, trending stocks, FX. Works wherever a smoothed trend view helps.

TIMEFRAME

Daily

ASSETS

Indices, trending stocks, FX

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
signal	candle color / color change	Long on bullish HA candles, exit on color flip.
confirmation	trend filter (e.g. MA)	Reduce range-bound flip-flopping.
exit	color change or doji-like HA candle	Indecision candles flag fading momentum.
price_for_orders	REAL OHLC, not HA values	HA open/close are averaged, not tradeable prices.

RISK & SIZING — SET TO YOUR OWN PLAN

Inherits the trend strategy's sizing and stops. The key correctness note: Heikin-Ashi open/close are smoothed averages, not real prices, so signals are read from HA candles but orders and stops must reference actual OHLC, otherwise fills are unrealistic. It lags like all smoothing.

WATCH OUT FOR

Mistaking smoothed HA prices for tradeable prices leads to unrealistic backtests; always execute on real OHLC. In ranges, HA candles still flip and whipsaw. The smoothing hides the actual high/low, so risk based on HA candles understates real range. It is a holding aid, not a precise timing tool.

EVIDENCE

Heikin-Ashi is a widely used trend-smoothing technique that helps discretionary and systematic traders stay with trends; its benefit is primarily noise reduction and holding discipline, and unlike Renko it preserves time, but its averaged prices are not tradeable and it shares the general lag of smoothing methods.

Copy into Claude Code — builds this strategy on your bot template

Build a Heikin-Ashi trend strategy as a BaseStrategy class. Compute Heikin-Ashi candles from real OHLC (HA close = average of OHLC; HA open = average of prior HA open and close; HA high/low from the extremes). Generate trend signals from HA candle color and color changes, with a trend-filter confirmation (default a moving average on real price) to reduce range whipsaw; exit on a color flip or an indecision (small-body, long-wick) HA candle. Crucially, read signals from HA candles but place all orders and stops at REAL OHLC prices, not HA values. Expose the signal mode, confirmation filter, and exit rule. Return diagnostics: HA candle color/body, confirmation state. Docstring MUST warn that HA prices are averaged and not tradeable (execute on real OHLC), that ranges still cause flips, and that it lags. Unit tests: HA candle computation, color-change signals, that order prices use real OHLC not HA values, trend-filter gating. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Point-and-Figure Breakout

Columns of Xs and Os that filter time and noise to flag clean breakouts.

HOW IT WORKS

Point-and-Figure (P&F) plots rising prices as columns of Xs and falling prices as columns of Os, adding a mark only when price moves a set box size, and switching columns only on a reversal of several boxes. Classic signals are breakouts above a prior column of Xs (double-top breakout) or below a prior column of Os. Like Renko, it ignores time and filters noise, emphasizing support/resistance and breakout levels.

WHERE THE EDGE COMES FROM

Noise and time filtering that highlights clean horizontal support/resistance and breakouts, reducing the false signals that clutter time-based charts.

WHEN TO USE IT

Breakout/trend trading on daily data where you want clean, objective breakout levels. A niche but long-established method, useful for defining breakout entries and price targets via P&F counting. Confirm with another filter; like all charting lenses the underlying move must be real.

BEST REGIME

Trending/breakout regimes

AVOID IN

Choppy ranges (repeated false breakouts), short intraday horizons

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily. The box-move construction ignores time, so intraday use is uncommon and noisier.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Stocks, ETFs, indices, commodities. Breakout-oriented instruments.

TIMEFRAME

Daily

ASSETS

Stocks, ETFs, indices, commodities

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
box_size	fixed or ATR-derived	Price move per box. Lock before testing (repaint risk like Renko).
reversal	3-box (classic)	Boxes needed to start a new column.
signal	double-top/bottom breakout	Break of a prior X/O column.
target	P&F vertical/horizontal count	Optional projected price target.
backtest_on	underlying prices	P&F prices are constructed; test on real prices.

RISK & SIZING — SET TO YOUR OWN PLAN

Position sizing by normal volatility rules. As with Renko, P&F is built from box moves and changing box/reversal settings repaints the chart, so lock settings before testing and backtest on real prices. Breakouts have a low win rate / good payoff, so trail and let winners run.

WATCH OUT FOR

Same repaint caution as Renko: box and reversal settings change the whole chart, and constructed P&F levels are not literal fill prices. False breakouts in ranges are the main cost. It is an objective lens for breakout levels, not a complete system; confirm and manage risk separately.

EVIDENCE

Point-and-Figure is one of the oldest charting methods with a long practitioner history for defining breakouts and support/resistance objectively; its edge, like other charting lenses, depends on the underlying breakout being real, and it shares Renko's settings-dependent repaint caveat.

Copy into Claude Code — builds this strategy on your bot template

```
Build a Point-and-Figure breakout strategy as a BaseStrategy class. Construct P&F columns from the underlying price series with a configurable box size (fixed or ATR-derived) and reversal amount (default 3-box), LOCKING the settings so the chart does not repaint. Generate signals on double-top breakouts (price breaks above a prior X column) for longs and double-bottom breakdowns for shorts if enabled; optionally compute a P&F count-based price target. Backtest fills must use real underlying prices, not constructed P&F levels. Expose box_size method, reversal, a shorts flag, and the target option. Return diagnostics: current column type, box size, last breakout level. Docstring must warn about the repaint caveat (lock settings before testing), that P&F levels are not literal fill prices, and false breakouts in ranges. Unit tests: column construction, double-top breakout signal, locked-settings behavior, fills on real prices. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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AC-04 Range Bars

Bars defined by a fixed price range rather than time, normalizing volatility per bar.

HOW IT WORKS

A range bar completes only when price moves a fixed amount (the range), regardless of time, so every bar represents the same price travel. In fast markets many bars print; in quiet markets few. This normalizes each bar's volatility and can make breakouts and momentum cleaner. Trade momentum/breakouts on the range-bar series, with the same lens-not-edge caveat as Renko.

WHERE THE EDGE COMES FROM

Per-bar volatility normalization: equal-range bars make momentum and breakout patterns more comparable across market speeds, reducing some time-based noise.

WHEN TO USE IT

Intraday or daily momentum/breakout trading where normalizing per-bar volatility helps. A niche tool; like Renko it filters noise but the signal must still be a real move, and it carries similar backtesting caveats around synthetic construction.

BEST REGIME

Volatile/trending conditions where many bars print

AVOID IN

Dead-quiet markets (almost no bars), and naive backtesting on constructed bars

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday or daily. Intraday range-bar strategies are time-of-day sensitive and cost-heavy; quiet periods starve the bars.

DAY / CALENDAR

Neutral.

ASSET CLASS

Liquid futures, FX, active stocks. Needs enough movement to print bars.

TIMEFRAME

Intraday or daily

ASSETS

Liquid futures, FX, active stocks

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
range_size	fixed or ATR-derived	Price travel per bar. Lock before testing.
signal	breakout/momentum on range bars	E.g. close beyond prior range bars.
confirmation	trend or volatility filter	Avoid chop.
stop	fixed number of ranges	Where the move is invalidated.
backtest_on		Constructed bars need careful, tick-aware testing.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
	tick/underlying prices	

RISK & SIZING — SET TO YOUR OWN PLAN

Normal volatility sizing. Like Renko and P&F, range bars are constructed, so accurate backtesting needs tick or underlying-price handling and locked range settings; testing naively on the constructed bars overstates accuracy. Intraday versions are cost-sensitive.

WATCH OUT FOR

Constructed-bar backtesting inaccuracy and settings-dependence (changing range size changes the bars). In quiet markets almost no bars form, starving the strategy of signals. Like all charting lenses, it does not create an edge where none exists; the underlying move must be real.

EVIDENCE

Range bars are an established alternative bar type used to normalize volatility per bar and clarify momentum, valued in futures/FX intraday trading; their benefit is noise/volatility normalization, and they share the synthetic-construction backtesting caveats of other non-time bars.

Copy into Claude Code — builds this strategy on your bot template

```
Build a range-bar momentum/breakout strategy as a BaseStrategy class. Construct range bars from the underlying (ideally tick or fine-grained) price series with a fixed or ATR-derived range size, LOCKED before testing. Generate breakout/momentum signals on the range-bar series (e.g. close beyond the prior N range bars) with a trend/volatility confirmation filter; stop after a fixed number of ranges against entry. Backtesting must handle the constructed nature carefully using underlying/tick prices, not assume clean fills on constructed bars. Expose range_size method, the breakout lookback, confirmation, and stop in ranges. Return diagnostics: current range size, bars formed recently, breakout level. Docstring must warn about constructed-bar backtesting caveats, locked settings, signal starvation in quiet markets, and that it is a lens not an edge. Unit tests: range-bar construction, breakout signal, locked-settings behavior, tick-aware fill handling. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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FAMILY 08 • 3 STRATEGIES

Microstructure & Intraday

Short-horizon strategies that exploit the structure of the trading day: the opening range, order-flow imbalance, gaps. Higher trade frequency, tighter stops, and far more sensitive to transaction costs and slippage than daily strategies.

MS-01 Opening-Range Breakout (ORB)

MS-02 Order-Flow Imbalance

MS-03 Gap-and-Go

Opening-Range Breakout (ORB)

Mark the first N minutes' high/low, then trade the breakout of that range.

HOW IT WORKS

After the open, define the opening range as the high and low of the first N minutes (e.g. 15, 30, or 60). A break above the range high goes long, a break below the range low goes short, with a stop at the opposite side of the range or a fraction of it. Exit at a target multiple or end of day. The opening range captures the session's initial balance, and a decisive break often sets the day's direction.

WHERE THE EDGE COMES FROM

The opening range reflects the day's initial supply/demand balance; a clean break signals which side won and tends to see follow-through, especially with a volume or volatility catalyst.

WHEN TO USE IT

Intraday on liquid instruments, strongest on days with a catalyst (news, gap, earnings elsewhere) that produces a directional open. Standalone day-trading strategy; add filters (only trade if the first bar has strong volume, or only on gap days) to avoid range-day chop.

BEST REGIME

Trend days, catalyst-driven sessions

AVOID IN

Quiet range days (the breakout fails and reverses)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Defined by time-of-day: the opening range is the first N minutes, and you flatten by close. The morning is the entire strategy.

DAY / CALENDAR

Catalyst days (CPI, FOMC, jobs, earnings clusters) produce the best trend-day breakouts; quiet mid-week sessions produce more failures.

ASSET CLASS

Index futures, liquid stocks, ETFs. Needs liquid intraday instruments.

TIMEFRAME

Intraday; flat by close

ASSETS

Index futures (ES, NQ), liquid stocks, ETFs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
range_minutes	15 / 30 / 60	Length of the opening range. Shorter = earlier, noisier.
entry	break of range high/low	Long above, short below.
stop	opposite side of range	Or a fraction of the range width.
target	R-multiple or measured move	E.g. 1-2x the range width.
filter	volume/volatility/gap	Trade only when a catalyst is present.

RISK & SIZING — SET TO YOUR OWN PLAN

Per-trade risk is the range width (or a fraction), so size by that distance to a fixed percent of equity. Higher frequency and intraday slippage mean smaller size and realistic cost modeling. Flat by close. A volume/volatility filter that skips quiet range days is the main way to cut losing breakouts.

WATCH OUT FOR

Range days produce false breakouts that immediately reverse (the classic 'fade the failed ORB' setup exists for a reason). Without a catalyst filter, ORB chops. Shorter opening ranges are noisier. Costs and slippage matter more than on daily strategies.

EVIDENCE

Opening-range breakout is a long-used, well-known intraday strategy (popularized by Crabel and others); it performs best on catalyst/trend days and poorly on quiet range days, so a volatility/volume filter and disciplined stops are standard. Edge is regime/day-type dependent.

Copy into Claude Code — builds this strategy on your bot template

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Build an opening-range breakout (ORB) strategy as a BaseStrategy class for intraday bars. Define the opening range as the high/low of the first range_minutes (default 30) of the session. Long on a break above the range high, short on a break below the range low; stop at the opposite side of the range (or a configurable fraction), target an R-multiple of the range width (default 1.5x), flat by close. Add a catalyst filter that only enables entries when opening-range volume or volatility exceeds a threshold (or only on gap days). Expose range_minutes, stop mode, target multiple, and the filter. Return diagnostics: range high/low, breakout side, range width, filter state. Docstring should note false breakouts on quiet range days, the need for the catalyst filter, intraday cost sensitivity, and flat-by-close. Unit tests: opening-range computation, breakout entries, stop placement, filter gating, forced close. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Order-Flow Imbalance

Trade short-term pressure from the imbalance between buying and selling volume.

HOW IT WORKS

Classify volume as buyer- or seller-initiated (e.g. trades at/above the ask vs at/below the bid, or via tick rule) and compute an imbalance over a short window. A strong buy imbalance signals upward pressure; a sell imbalance, downward. Enter in the direction of the imbalance for a short hold, exit on mean reversion of the imbalance or a quick target. This is a microstructure/flow strategy needing trade-level or tick data.

WHERE THE EDGE COMES FROM

Short-horizon price impact: aggressive order flow moves price, and persistent imbalance predicts continued short-term movement before liquidity refills.

WHEN TO USE IT

Very short-term intraday on liquid instruments, for members with access to trade/tick data and low-latency execution. Advanced; highly sensitive to costs and data quality. Best on liquid futures where flow classification is cleaner.

BEST REGIME

Liquid, active markets with clear flow

AVOID IN

Illiquid names, wide spreads, and anyone without proper tick data / fast execution

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Seconds to minutes, intraday. Highly time-of-day sensitive: flow is richest near the open and close, thin midday.

DAY / CALENDAR

Neutral on day-of-week; sensitive to event-driven liquidity.

ASSET CLASS

Liquid futures (ES, NQ) and highly liquid stocks. Needs trade/tick data; useless on thin names.

TIMEFRAME

Seconds to minutes; intraday

ASSETS

Liquid futures (ES, NQ), highly liquid stocks

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
classification	tick rule / at-bid-vs-ask	How to label buy vs sell volume.
window	short (seconds to a few min)	Imbalance lookback.
threshold	imbalance magnitude	How lopsided before acting.
hold	short	Exit quickly on imbalance reversion or target.
data	trade/tick level required	Bar data is too coarse for this.

RISK & SIZING — SET TO YOUR OWN PLAN

Smallest size and tightest stops in the library because horizon is shortest and costs bite hardest. Realistic slippage and fee modeling are make-or-break; a strategy that looks profitable on mid-prices can be a loser after costs. Requires quality tick data; bar-based approximations are unreliable. Many members should treat this as study-only.

WATCH OUT FOR

Transaction costs and slippage dominate at this horizon and routinely turn a 'profitable' signal into a loss; mid-price backtests lie. Needs real tick data and fast execution that most retail setups lack. Flow classification is imperfect. Easy to overfit on microstructure noise.

EVIDENCE

Order-flow imbalance is a well-studied microstructure predictor of short-horizon returns and is used in professional execution and HFT-adjacent strategies; its accessibility to retail is limited by data, latency, and cost sensitivity, so it is presented here as advanced and cost-critical.

Copy into Claude Code — builds this strategy on your bot template

Build an order-flow imbalance strategy as a BaseStrategy class for very-short-horizon intraday trading, assuming trade/tick data is available. Classify volume as buyer- or seller-initiated (tick rule or at-bid-vs-ask) and compute an imbalance over a short window (default a few minutes). Enter in the direction of a strong imbalance (above a threshold) for a short hold; exit on imbalance reversion or a small target. Require realistic transaction-cost and slippage models and compute net (not mid-price) expectations. Expose the classification method, window, threshold, and hold rule. Return diagnostics: current imbalance, classification counts, net-of-cost edge estimate. Docstring must state this needs tick data and fast execution, that costs dominate and mid-price backtests are misleading, and that it is advanced/study-level for most retail setups. Unit tests: volume classification, imbalance computation, threshold entries, cost-adjusted edge calculation. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Gap-and-Go

Trade *WITH* a strong, news-driven opening gap that continues rather than fills.

HOW IT WORKS

The opposite of gap fading: when a stock gaps on real news/earnings with strong premarket volume, trade in the gap's direction expecting continuation as the move attracts momentum buyers. Entry on a break of the premarket high (for a gap up) after the open, stop below the opening range or a key level, exit on a target or trailing stop. The news catalyst is the whole point.

WHERE THE EDGE COMES FROM

Momentum and continuation on genuine catalysts: a strong, high-volume news gap reflects real new information and often sees follow-through as participants react over the session (related to PEAD intraday).

WHEN TO USE IT

Intraday momentum day-trading on stocks gapping on real catalysts with heavy premarket volume. Standalone. The catalyst and volume filters distinguish it from fading: gap-and-go trades news gaps, gap-fade trades emotional non-news gaps.

BEST REGIME

Strong catalyst, high relative volume, trending open

AVOID IN

Low-volume or no-news gaps (those tend to fill, the fade trade)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Intraday, anchored to the open. Enter on the post-open continuation, flatten by close. A morning strategy.

DAY / CALENDAR

Driven by catalysts (earnings, news) which cluster rather than follow weekdays.

ASSET CLASS

Liquid stocks with news catalysts, often more volatile names. Single-name intraday momentum.

TIMEFRAME

Intraday; same-day

ASSETS

Liquid stocks with news catalysts (often more volatile names)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
catalyst	real news/earnings required	No catalyst = expect a fade instead.
rel_volume	high premarket/relative volume	Confirms real participation.
entry	break of premarket high/open range	Enter on continuation confirmation.
stop	below opening range / key level	Tight, defined.
exit		Ride continuation; do not overstay.

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
	trailing stop or target	

RISK & SIZING — SET TO YOUR OWN PLAN

Tight, defined stops because these are volatile names; size by the stop distance to a fixed percent and keep size modest given the volatility. Flat by close (or defined swing rules). The catalyst + relative-volume filter is a risk control: trading gap-and-go on a no-news gap is just fading in disguise and likely to lose.

WATCH OUT FOR

Without a real catalyst, the gap tends to fill (you are on the wrong side, the fade trade). High-volatility names can reverse hard; tight stops and small size are essential. Chasing an extended move late gives poor risk/reward. Spreads on fast-moving names add cost.

EVIDENCE

Gap-and-go / momentum continuation on catalyst gaps is a staple of intraday momentum trading and is the intraday cousin of post-earnings drift; it works on genuine high-volume news gaps and fails on low-volume non-news gaps, which is why catalyst and volume filters are central.

Copy into Claude Code — builds this strategy on your bot template

```
Build a gap-and-go momentum strategy as a BaseStrategy class for intraday trading. Require a real catalyst flag (news/earnings) and a high relative-volume condition for the symbol; if either is missing, do not trade (that is fade territory). For a qualifying gap up, enter long on a break of the premarket high or opening-range high after the open; mirror for gap downs if shorting is enabled. Stop below the opening range or a key level; exit on a trailing stop or target; flat by close. Expose the required catalyst and relative-volume inputs, entry trigger, stop mode, and exit. Return diagnostics: gap size, relative volume, catalyst flag, entry level. Docstring must contrast with gap-fade (this trades WITH news gaps), stress the catalyst/volume filter as a risk control, and warn about volatility and chasing late. Unit tests: no trade without catalyst/volume, continuation entry trigger, stop placement, forced close. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Seasonality & Macro

Calendar-season and macro-driven tilts: sell-in-May, the turn-of-year effect, sector seasonality, carry. Weak as standalone signals, useful as overlays or tilts on top of a real strategy. Treat the documented effects as small and partly decayed.

SE-01 Sell-in-May / Halloween Effect

SE-02 Santa Claus Rally / Turn-of-Year

SE-03 Cross-Asset Carry

SE-04 Day-of-Week / Turn-of-Week Effect

Sell-in-May / Halloween Effect

Equities have historically returned more in November-April than May-October.

HOW IT WORKS

The 'Halloween indicator' holds equities from roughly November through April and reduces exposure (to cash or bonds) May through October, based on the long-documented seasonal pattern that winter-half returns have exceeded summer-half returns in many markets. The simplest version switches a broad index ETF on/off by calendar; refinements add a trend or momentum confirmation.

WHERE THE EDGE COMES FROM

A persistent, widely documented seasonal pattern in equity returns (the summer-underperformance effect), attributed variously to seasonal risk appetite, liquidity, and behavioral factors.

WHEN TO USE IT

As a low-effort seasonal tilt/overlay, not a precise timing system. Better combined with a trend filter (stay invested in summer if the trend is strongly up) than applied blindly. Useful as a teaching example of calendar seasonality.

BEST REGIME

Normal regimes; the seasonal tilt is a small edge

AVOID IN

Letting the calendar override a strong trend in either direction

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Semiannual switch on the close. No intraday or clock component.

DAY / CALENDAR

Driven by the calendar month (the Nov-Apr vs May-Oct split), not weekdays.

ASSET CLASS

Broad equity-index ETFs. An equity-seasonality overlay.

TIMEFRAME

Semiannual switch

ASSETS

Broad equity-index ETFs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
in_months	Nov-Apr invested	The 'winter' half.
out_months	May-Oct reduced/cash	The 'summer' half.
confirmation	optional trend filter	Override the calendar if the trend strongly disagrees.
out_allocation	cash or bonds	Where to sit in summer.
size	tilt, not all-in	Treat as an overlay.

RISK & SIZING — SET TO YOUR OWN PLAN

Small, overlay-level allocation; this is a tilt, not a standalone engine. Combining with a trend filter avoids the worst outcome (sitting out a strong summer rally or staying long into a summer crash). Do not lever a seasonal effect.

WATCH OUT FOR

It is a small, well-known, partly-decayed effect with plenty of years where summer beats winter; blind application has long unrewarded stretches. Over-fitting exact switch dates is easy. It will not protect against a winter bear market.

EVIDENCE

The sell-in-May / Halloween effect is one of the most studied seasonal anomalies, documented across many countries and long samples with persistent (if modest) summer underperformance, but its magnitude is small, it has exceptions, and it is best used as an overlay rather than a sole strategy.

Copy into Claude Code — builds this strategy on your bot template

Build a sell-in-May / Halloween-effect overlay as a BaseStrategy class for a broad index ETF. Signal invested during in_months (default November-April) and reduced/cash during out_months (default May-October), with the out-of-market allocation configurable to cash or bonds. Add an optional trend-filter override that keeps you invested in summer when the trend is strongly up (and out in winter if strongly down). Expose the month ranges, out-allocation, and the trend override. Return diagnostics: current half (winter/summer), trend-override state. Docstring should note this is a small, partly-decayed seasonal tilt best used as an overlay with a trend filter, not a precise timer, and that it will not protect against a winter bear. Unit tests: month-based in/out signal, out-allocation, trend override behavior. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Santa Claus Rally / Turn-of-Year

A documented tendency for equities to rise in the last days of December into early January.

HOW IT WORKS

The 'Santa Claus rally' refers to a historical tendency for equity indices to gain over the final five trading days of December and the first two of January, often linked to holiday-season optimism, light volume, tax considerations, and bonus flows. The simple version holds a broad index over that specific window. Related turn-of-year small-cap effects (the 'January effect') overlap.

WHERE THE EDGE COMES FROM

Seasonal flows and behavior around year-end and the new year: tax-driven selling pressure abates, holiday optimism and reinvestment flows push prices up in a thin-liquidity window.

WHEN TO USE IT

As a tiny seasonal overlay around year-end, not a standalone system. Mainly illustrative of turn-of-year seasonality. Best layered on broad index exposure for the specific window.

BEST REGIME

Normal year-end conditions

AVOID IN

Year-end during an acute sell-off (seasonality does not override a crisis)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily; hold across the year-end window. Defined in trading days around the turn of the year.

DAY / CALENDAR

Entirely calendar-driven (last five December sessions plus first two January sessions).

ASSET CLASS

Broad equity-index ETFs, with an optional small-cap tilt for the January effect.

TIMEFRAME

~7 trading days around year-end

ASSETS

Broad equity-index ETFs; small-cap tilt for the January effect

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
window	last 5 of Dec + first 2 of Jan	The classic Santa Claus window.
instrument	broad index (or small caps)	Small caps for the related January effect.
size	small overlay	One short window per year.
regime_gate	stand down in a crisis	Seasonality is not crash protection.

RISK & SIZING — SET TO YOUR OWN PLAN

Minimal size; one short window a year means tiny sample and high variance, so do not over-allocate. A regime gate avoids being long into the window during an active crisis. Strictly an overlay.

WATCH OUT FOR

One occurrence per year is statistically thin; a few bad years dominate the record. The effect is small and well known. The related January small-cap effect has weakened over time. Not crash protection.

EVIDENCE

The Santa Claus rally and turn-of-year effects are documented seasonal patterns with a long (if noisy) history; the related January small-cap effect is well studied and has attenuated over time. All are small, thin-sample effects suited only to overlay-level use.

Copy into Claude Code — builds this strategy on your bot template

Build a Santa Claus rally / turn-of-year overlay as a BaseStrategy class for a broad index ETF (with an optional small-cap tilt for the January effect). Using a trading calendar, signal long during the window of the last five trading days of December plus the first two of January, flat otherwise. Include a regime gate that stands down during an active crisis. Keep size small. Expose the window definition, instrument, and regime gate. Return diagnostics: in-window state, days remaining in window. Docstring should note the thin once-a-year sample, the small and well-known nature of the effect, the attenuated January small-cap effect, and that it is not crash protection. Unit tests: window detection across the year boundary, flat outside the window, regime stand-down. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Cross-Asset Carry

Hold high-yielding assets funded by low-yielding ones; collect the yield differential.

HOW IT WORKS

Carry strategies go long assets with high 'carry' (yield, roll, or interest) and short or underweight low-carry assets, earning the differential while the price stays put. Classic examples: currency carry (long high-rate currencies, short low-rate), commodity carry (backwardated vs contango futures), and bond carry (rolling down the yield curve). It earns a steady premium that can reverse sharply in risk-off events.

WHERE THE EDGE COMES FROM

The carry premium: investors are compensated for bearing the risk that high-carry assets sometimes crash, a documented cross-asset risk premium related to but distinct from momentum and value.

WHEN TO USE IT

Advanced, cross-asset, for members comfortable with futures/FX. Best as a diversifying sleeve combined with momentum (carry and momentum are complementary). Must be risk-managed for the sharp, correlated drawdowns carry suffers in crises.

BEST REGIME

Calm, risk-on regimes

AVOID IN

Risk-off / crisis regimes (carry trades unwind violently and in correlation)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly rebalance on the close.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Cross-asset: currencies, commodity futures, bonds. A cross-asset carry tilt, not single equities.

TIMEFRAME

Monthly rebalance

ASSETS

Currencies, commodity futures, bonds (cross-asset)

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
carry_measure	yield / roll / rate differential	Defined per asset class.
universe	cross-asset or within one class	FX carry, commodity carry, etc.
long_short	long high-carry, short low-carry	Market-neutral-ish within the class.
rebalance	monthly	Carry ranks change slowly.
risk_overlay	vol target + crisis gate	Carry needs drawdown protection.

RISK & SIZING — SET TO YOUR OWN PLAN

Carry's return profile is steady gains punctuated by sharp crash-correlated losses (the 'carry crash'), so a volatility-target overlay and a risk-off/crisis gate are essential, and sizing must respect the fat left tail. Diversify across asset classes; do not concentrate carry in one market. Combining with momentum dampens the drawdowns.

WATCH OUT FOR

Carry trades unwind violently and together in risk-off episodes (the classic FX carry crash), so the smooth equity curve hides real tail risk. Funding/borrow and roll costs eat the differential. It requires cross-asset data and futures/FX access. Naive, unhedged carry is dangerous.

EVIDENCE

Carry is a well-documented cross-asset risk premium (currencies, commodities, bonds, equities) with persistent average returns and a characteristic left-skewed, crash-prone profile; it is a staple of systematic multi-asset programs, typically combined with momentum and run with explicit drawdown controls.

Copy into Claude Code — builds this strategy on your bot template

```
Build a cross-asset carry strategy as a BaseStrategy-style class over a configurable cross-asset universe (e.g. FX, commodity futures, or bonds). Define a carry measure per asset (yield, roll, or rate differential), rank assets monthly, go long high-carry and short low-carry within the class. Add a mandatory risk overlay: volatility targeting plus a risk-off/crisis gate that cuts exposure in high-vol regimes. Optionally blend with a momentum signal to dampen drawdowns. Expose the carry measure, universe, long_short construction, rebalance frequency, and the risk overlay. Return diagnostics: carry ranks, positions, vol-scaled weights, regime state. Docstring must state carry has a crash-prone left tail (the carry crash), needs the vol target and crisis gate, and benefits from a momentum blend. Unit tests: carry ranking, long-short construction, vol-target scaling, crisis-gate exposure cut. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Day-of-Week / Turn-of-Week Effect

Historical weekday return patterns in equities, used as a small timing tilt rather than a system.

HOW IT WORKS

Equity returns have shown weekday patterns over long samples, the classic 'weekend effect' (historically weak Mondays) and turn-of-week tendencies. The simplest use is a tilt: lean exposure toward historically stronger parts of the week or avoid putting on fresh risk at historically weak times, on a broad index. It is a tiny, partly-decayed effect best layered on a real strategy, not traded alone.

WHERE THE EDGE COMES FROM

Calendar/behavioral patterns in when returns accrue across the week, attributed to weekend information flow, settlement timing, and order-flow patterns. The original weekend effect has weakened and at times reversed.

WHEN TO USE IT

As a minor timing overlay deciding when other strategies put on or hold risk, on broad equity indices. Useful mainly as a teaching example of day-of-week seasonality. Do not allocate meaningfully to it standalone.

BEST REGIME

Normal regimes; a marginal tilt

AVOID IN

Letting a weekday rule override a real signal or a crash

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily, with intra-week timing. Acts on which day/session to add or hold risk; not an intraday clock strategy.

DAY / CALENDAR

This is the entire basis of the strategy.

ASSET CLASS

Broad equity-index ETFs. An index-level seasonal tilt.

TIMEFRAME

Daily; intra-week timing

ASSETS

Broad equity-index ETFs

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
pattern_source	estimate from your own data	Patterns drift and reverse; re-estimate, do not assume the textbook version.
use	tilt entry/hold timing	Lean toward stronger windows, avoid weak ones for new risk.
instrument	broad index	An index-level effect.
size	minimal tilt	One of the weakest effects in the library.
regime_gate	stand down in stress	Weekday patterns do not survive a crash.

RISK & SIZING — SET TO YOUR OWN PLAN

Minimal size; this is a marginal tilt, not a return engine. Re-estimate the pattern on current data rather than trusting the historical weekend effect, which has decayed and at times flipped. Use it to time when other strategies act, and never let it override a real signal or a risk-off regime.

WATCH OUT FOR

The original weekend/Monday effect has weakened and sometimes reversed, so the textbook version can be wrong out-of-sample. It is easy to data-mine spurious weekday patterns. The effect is tiny and not worth standalone capital. It will not help in a sell-off.

EVIDENCE

Day-of-week effects (notably the weekend effect) are among the most studied calendar anomalies, with strong historical evidence that has materially weakened and in places reversed in recent decades; they are best treated as a small, possibly-unstable timing tilt rather than a tradeable standalone edge.

Copy into Claude Code — builds this strategy on your bot template

```
Build a day-of-week / turn-of-week tilt as a class on my BaseStrategy interface for a broad index ETF. Rather than hardcoding the textbook weekend effect, estimate weekday return tendencies from supplied historical data (with a clear note that patterns drift and must be re-estimated, not assumed), then use the result as a small timing tilt: lean toward historically stronger windows and avoid putting on fresh risk in weaker ones. Include a regime gate that stands the tilt down in stress. Expose the pattern-estimation window, the tilt use, the instrument, and the regime gate. IMPORTANT: keep the allocation/size a parameter with a comment that this is a marginal tilt to be sized minimally per my own rules, and that it should not override a real signal. Return diagnostics: estimated weekday tendencies, current-day tilt, regime state. Docstring should state the weekend effect has decayed and sometimes reversed, the risk of data-mining weekday patterns, and that it is overlay-only. Unit tests: pattern estimation from data, tilt application by weekday, regime stand-down, size as a minimal parameter. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Regime & Allocation

Not signals at all, but rules for how much to hold and when. Regime detection (your HMM), volatility targeting, and risk parity decide gross exposure and position size. These wrap around the signal strategies and usually matter more for returns than the signal itself.

RA-01 HMM Regime Allocation

RA-02 Volatility-Regime Switching

RA-03 Risk-Parity Allocation

RA-04 Fractional-Kelly Position Sizing

RA-01

HMM Regime Allocation

Use the Hidden Markov Model regime to set gross exposure, no directional signal required.

HOW IT WORKS

Rather than generating buy/sell signals, this uses your HMM regime classification to decide how much risk to hold. In bull/low-vol regimes, run higher gross exposure to your core holdings; in bear/crash regimes, cut exposure or move to cash. It is allocation by regime: the regime engine is the only input, and it gates how aggressive every other strategy is allowed to be.

WHERE THE EDGE COMES FROM

Regime persistence and the strong difference in risk-adjusted returns across regimes: being more exposed in benign regimes and defensive in adverse ones improves risk-adjusted returns even with no stock-picking.

WHEN TO USE IT

As the top-level risk dial for your whole system, and as the gate that turns the other strategies in this library on or off by regime. This is the connective tissue between the library and your regime-trading bot guides.

BEST REGIME

Adds value whenever regimes persist

AVOID IN

Choppy regime transitions where the classification flips (use probability smoothing)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily check, slow exposure changes.
Not a timing tool.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Any core portfolio or the whole book.
Asset-agnostic; it sets exposure for everything below it.

TIMEFRAME

Daily check, slow changes

ASSETS

Any core portfolio / the whole book

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
regime_source	your HMM (filtered, past-only)	Use filtering, never look-ahead Viterbi smoothing.
exposure_map	regime -> gross exposure	E.g. bull 100%, range 70%, bear 30%, crash 0-20%.
smoothing	probability-weighted	Blend on regime probabilities to avoid flip-flop.
min_dwell	avoid one-day flips	Require persistence before big exposure changes.
apply_to	all sub-strategies	Gate each strategy's allowed size by regime.

RISK & SIZING — SET TO YOUR OWN PLAN

This IS the master risk control. Use probability-weighted exposure (not hard switching) so a borderline regime does not cause a jarring all-in/all-out move, and require some persistence before large changes. The single most important correctness rule: the regime must be computed with filtering (past-only), never look-ahead smoothing, or the whole thing leaks future information.

WATCH OUT FOR

Look-ahead leakage is the cardinal sin: using the smoothed (Viterbi) regime makes backtests glorious and live trading disappointing. Hard regime switching causes whipsaw; blend on probabilities. Over-frequent reclassification churns exposure. The regime model itself can be wrong, so keep exposure changes gradual.

EVIDENCE

Regime-based allocation using HMMs and similar models is a well-established approach to improving risk-adjusted returns by conditioning exposure on market state; its real-world value depends entirely on out-of-sample, look-ahead-free regime inference, which is why filtering (not smoothing) and probability blending are emphasized.

Copy into Claude Code — builds this strategy on your bot template

```
Build an HMM regime-allocation overlay as a BaseStrategy-style class that sets gross exposure rather than direction. Take the filtered (past-only) regime and its probability vector as input. Map regimes to target gross exposure via a configurable table (e.g. bull 100 percent, range 70, bear 30, crash 10). Compute the actual exposure as a probability-weighted blend across regimes (not a hard switch), and require a minimum dwell/persistence before large exposure changes. Provide a method other strategies call to learn their regime-allowed size. Expose the regime source, exposure map, smoothing, and min-dwell. Return diagnostics: regime, probabilities, blended exposure. Docstring MUST stress: use filtering not look-ahead Viterbi, blend on probabilities to avoid whipsaw, change exposure gradually, and that this is the master risk dial gating all other strategies. Unit tests: exposure mapping, probability-weighted blending, min-dwell preventing one-day flips, exposing regime-allowed size to sub-strategies. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Volatility-Regime Switching

Switch strategy style and exposure based on whether volatility is high or low.

HOW IT WORKS

Classify the volatility environment (e.g. realized vol above/below a threshold, or VIX level/percentile) and switch behavior accordingly: in low-vol regimes favor trend/momentum and run higher exposure; in high-vol regimes favor mean reversion or defense and cut exposure. It is a simpler, single-variable cousin of full HMM regime detection, using volatility as the regime proxy.

WHERE THE EDGE COMES FROM

Volatility regimes are persistent and predict which styles work: trend tends to do better in calm trending markets, reversion in choppy high-vol markets, so matching style to vol regime improves results.

WHEN TO USE IT

As a lightweight regime gate for members who want something simpler than a full HMM, or as a second confirming regime input alongside the HMM. Good for switching which sub-strategies are active and scaling exposure.

BEST REGIME

Whenever volatility regimes persist

AVOID IN

Rapid vol regime changes near the threshold (add hysteresis)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Daily. Sets style and exposure, not entry timing.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Whole-book or strategy-selection level. Asset-agnostic.

TIMEFRAME

Daily

ASSETS

Whole book / strategy selection

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
vol_measure	realized vol or VIX percentile	The regime variable.
threshold	e.g. VIX 20, or vol median	Splits high vs low regimes (consider multiple bands).
low_vol_style	trend/momentum, higher exposure	What to favor when calm.
high_vol_style	reversion/defense, lower exposure	What to favor when stressed.
hysteresis	buffer around threshold	Avoid flip-flopping right at the line.

RISK & SIZING — SET TO YOUR OWN PLAN

A regime/allocation control: it scales exposure and selects style. Add hysteresis (different thresholds for switching up vs down) so price hovering at the threshold does not cause constant switching. Like all regime tools, keep changes gradual and avoid acting on a single day's reading.

WATCH OUT FOR

Whipsaw right at the threshold if there is no hysteresis. A single vol variable is cruder than a full regime model and can misclassify. Realized vol lags; VIX is forward-looking but noisy. Treat it as one input, ideally confirming the HMM rather than fighting it.

EVIDENCE

Volatility-regime conditioning is well supported: style efficacy and risk-adjusted returns differ markedly between high- and low-volatility states, and switching exposure/style by vol regime is a standard, effective risk-management technique, simpler than but complementary to HMM regime detection.

Copy into Claude Code — builds this strategy on your bot template

Build a volatility-regime switching overlay as a BaseStrategy-style class. Classify the volatility environment using a configurable measure (realized vol or VIX percentile) and threshold(s), with hysteresis (separate up/down switch levels) to avoid flip-flopping at the line. In low-vol regimes, favor trend/momentum sub-strategies and higher gross exposure; in high-vol regimes, favor mean-reversion/defensive sub-strategies and lower exposure. Provide methods to report the current vol regime and the implied style/exposure. Expose the vol measure, threshold(s), hysteresis buffer, and the style/exposure maps. Return diagnostics: vol measure value, regime, exposure, active style. Docstring should note hysteresis prevents threshold whipsaw, a single vol variable is cruder than an HMM, and it is best as a confirming input. Unit tests: regime classification with hysteresis, style/exposure mapping, no flip-flop within the buffer. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.

Risk-Parity Allocation

Size positions so each contributes equal risk, not equal dollars.

HOW IT WORKS

Instead of equal-dollar or cap-weighted allocation, risk parity sizes each asset so its risk contribution to the portfolio is equal. Low-volatility assets (bonds) get larger weights, high-volatility assets (stocks) smaller, often with modest leverage to reach a target return. The goal is a more balanced, diversified risk exposure than a stock-heavy 60/40, which is dominated by equity risk.

WHERE THE EDGE COMES FROM

Diversification efficiency: equal-dollar portfolios concentrate risk in the most volatile asset, while equalizing risk contributions produces better-diversified, more stable risk-adjusted returns across regimes.

WHEN TO USE IT

As a portfolio-construction method for a multi-asset core, or to allocate across the strategies in this library by their risk contribution. A foundational allocation technique that pairs naturally with volatility targeting.

BEST REGIME

Diversified, multi-asset contexts across regimes

AVOID IN

Crises where correlations spike to 1 and diversification temporarily fails

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Monthly or quarterly rebalance on the close.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Multi-asset (stocks, bonds, commodities) or across the library's strategies. A portfolio-construction layer.

TIMEFRAME

Monthly or quarterly rebalance

ASSETS

Multi-asset (stocks, bonds, commodities) or across strategies

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>assets_or_strategies</code>	multi-asset or multi-strategy	Apply across asset classes or across the library's strategies.
<code>risk_measure</code>	volatility (and optionally correlation)	Inverse-vol is the simple version; full risk parity uses the covariance matrix.
<code>leverage</code>	optional, capped	To reach a target return; cap it hard.
<code>rebalance</code>	quarterly	Weights drift slowly.
<code>correlation_aware</code>	optional	Full risk parity accounts for correlations, not just vols.

RISK & SIZING — SET TO YOUR OWN PLAN

A construction method, but leverage is the danger: reaching target returns often implies leverage, which must be hard-capped, because in a correlation-spike crisis a levered risk-parity book can draw down sharply (as in 2020/2022 episodes). Diversification fails exactly when you need it most, so respect a leverage cap and a crisis gate.

WATCH OUT FOR

Correlations spiking to 1 in crises breaks the diversification assumption and levered risk parity can suffer; the smooth historical ride hides this. Inverse-vol-only versions ignore correlations. Bond-heavy weightings are rate-sensitive. Leverage turns a stable method into a risky one if uncapped.

EVIDENCE

Risk parity is a well-documented, widely used allocation framework (popularized by large multi-asset funds) that improves diversification of risk versus equal-dollar/60-40 portfolios; its weakness is correlation breakdown and leverage sensitivity in crises, which is why capped leverage and risk overlays are standard.

Copy into Claude Code — builds this strategy on your bot template

```
Build a risk-parity allocation component as a BaseStrategy-style class that can allocate across either a multi-asset set or across the library's strategies. Compute each component's volatility (inverse-vol weighting for the simple version) or use the covariance matrix for full risk parity if correlation_aware is true, setting weights so risk contributions are equalized. Support optional, hard-capped leverage to reach a target risk/return, and rebalance on a configurable cadence (default quarterly). Expose the component set, risk measure, leverage cap, rebalance frequency, and the correlation_aware flag. Return diagnostics: per-component vol/risk contribution, weights, total leverage. Docstring MUST warn that correlations spike in crises (diversification can fail), leverage must be capped, and inverse-vol ignores correlations. Unit tests: inverse-vol weighting, risk-contribution equalization, leverage cap enforcement, rebalance cadence. Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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Fractional-Kelly Position Sizing

Size positions by a fraction of the Kelly criterion to balance growth against drawdown.

HOW IT WORKS

The Kelly criterion gives the bet size that maximizes long-run growth given a strategy's edge and odds. Full Kelly is too aggressive in practice (estimation error and fat tails make it ruinously volatile), so practitioners use a fraction, commonly a half or quarter Kelly. This is a sizing layer: estimate your strategy's edge and variance, compute the Kelly fraction, scale it down by your chosen fraction, and cap it. It decides how much, not what.

WHERE THE EDGE COMES FROM

Not a signal but a growth-optimal sizing framework: Kelly maximizes long-run compounding for a known edge, and using a fraction trades a little growth for a large reduction in volatility and drawdown.

WHEN TO USE IT

As a sizing layer on any strategy with an estimable edge and win/loss distribution. Best combined with volatility targeting and your regime exposure dial. For members who want a principled alternative to fixed-percent sizing.

BEST REGIME

Any strategy with a stable, estimable edge

AVOID IN

Strategies whose edge is unstable or hard to estimate (Kelly is then unreliable)

WHEN TO RUN IT: TIME, DAY & ASSET FIT

TIME OF DAY

Recompute as edge estimates update (e.g. on rebalance or periodically). Not a timing tool.

DAY / CALENDAR

Irrelevant.

ASSET CLASS

Asset-agnostic. It sizes any strategy in any class that has an estimable edge.

TIMEFRAME

Recompute as edge estimates update

ASSETS

Any

CONFIGURATION

PARAMETER	DEFAULT (SET YOUR OWN)	NOTES
<code>kelly_fraction</code>	0.25 to 0.5	Fraction of full Kelly. Lower is safer; full Kelly is too volatile.
<code>edge_estimate</code>	from backtest / rolling stats	Win rate and payoff, or mean/variance of returns.
<code>max_position_cap</code>	hard cap	Never let Kelly exceed a sane per-position limit.
<code>recompute</code>	periodically	Update as edge estimates change; avoid overreacting to noise.
<code>combine_with</code>	vol targeting + regime dial	Layer with other risk controls.

RISK & SIZING — SET TO YOUR OWN PLAN

This is a sizing/risk framework, and the danger is over-trusting the edge estimate: Kelly is exquisitely sensitive to the assumed edge, and an overstated edge produces dangerous oversizing, which is why a fraction (quarter to half) and a hard position cap are essential. Recompute edge estimates conservatively and never use full Kelly on a real, uncertain strategy.

WATCH OUT FOR

Garbage-in: if your edge estimate is too high (easy with an overfit backtest), even fractional Kelly oversizes and the drawdowns are brutal. Full Kelly is famously too volatile for real trading. Edge and variance drift, so a stale estimate misleads. It assumes you can estimate the edge at all, which many strategies cannot.

EVIDENCE

The Kelly criterion is a well-established growth-optimal sizing result, and fractional-Kelly sizing is widely used by practitioners precisely because full Kelly's volatility is impractical; its real-world value is entirely contingent on a realistic edge estimate, hence the emphasis on fractions and caps.

Copy into Claude Code — builds this strategy on your bot template

```
Build a fractional-Kelly position-sizing layer as a reusable component that wraps any strategy on my BaseStrategy interface. Estimate the strategy's edge from supplied statistics (win rate and payoff, or mean and variance of returns), compute the full-Kelly fraction, scale it by a configurable kelly_fraction (default 0.25), and clamp to a hard per-position cap. Recompute on a configurable cadence and avoid overreacting to noisy estimates. Expose kelly_fraction, the edge-estimate inputs, max_position_cap, and recompute cadence. IMPORTANT: this component sets size, so make the cap and fraction prominent parameters and state clearly in the docstring that Kelly is highly sensitive to the edge estimate, that an overstated edge causes dangerous oversizing, and that full Kelly should never be used on a real uncertain strategy. Return diagnostics: estimated edge, raw Kelly, fractional Kelly, capped size. Unit tests: Kelly computation from edge stats, fractional scaling, hard cap enforcement, behavior when the edge estimate is zero or negative (size to zero). Finally, do not treat the defaults in this prompt as final. Expose the key parameters (sizing, stop, max loss per trade, and the main strategy settings) as configurable inputs, and add a short comment block telling me to set sizing, risk, and the regime, timeframe, day, and asset-class fit to my own plan before running. Note clearly that this is a starting template to configure and backtest on my own data, not a plug-and-play system.
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